Journal of Retailing and Consumer Services 85 (2025) 104294

Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser

Digital daydreams: Exploring consumer motivations for engaging with the metaverse

Sigma Soni [a,1,*], Parvinder Arora [b,1], Dharun Kasilingam [c,1], Varsha Jain [d]

[a] Symbiosis Institute of Media & Communication, Lavale, Mulshi, Pune, Maharashtra, 412115, India
[b] International Management Institute, B-10, Qutab Institutional Area, Tara Crescent, New Delhi, 110016, India
[c] Indian Institute of Management Kozhikode, IIMK Campus PO, Kozhikode, 673 570, Kerala, India
[d] MICA, Shela, Ahmedabad, 380 058, Gujarat, India

ARTICLE INFO

Keywords:
Metaverse
Behavioral reasoning theory
Technology readiness
Customer engagement
Mixed methods
Technology adoption

ABSTRACT

The metaverse is rapidly emerging as a transformative digital technology with significant implications for consumer behavior and marketing strategies. However, there is insufficient knowledge about the factors that can motivate or prevent consumers from adopting the metaverse. Addressing this gap, this study employs Behavioral Reasoning Theory (BRT) and Technology Readiness frameworks to investigate the dual forces of motivation and resistance that shape metaverse adoption. This research used a mixed-methods approach, which consisted of in-depth interviews with 48 industry experts to determine the 'reasons for' and 'reasons against' metaverse adoption, which were then validated through a survey of 771 consumers. The findings reveal that positive motivators like haptic imagery, convenience, trend affinity, and relative advantage significantly enhance consumer attitudes. Meanwhile, barriers like privacy concerns, financial concerns, risk of malfunctions, and perceived performance risks are negative influencers. Results also show that technology readiness is a crucial determinant, with optimism and innovativeness promoting adoption and discomfort and insecurity deterring it. This study contributes to the metaverse adoption literature by examining consumer attitudes' role in shaping adoption intentions and engagement, an area that has received relatively limited attention. The findings offer actionable strategies for practitioners to navigate consumer hesitations and unlock the metaverse's potential as a next-generation interaction, innovation, and commerce platform.

1. Introduction

The metaverse signifies an evolution in digital consumer experiences by combining virtual and augmented realities to enhance social interactions, commerce, and entertainment (Kim, 2021). It is a collective virtual space that amalgamates virtual reality (VR), augmented reality (AR), blockchain technology, artificial intelligence (AI), and digital payment systems (NFTs and cryptocurrencies) to deliver immersive and interactive encounters (GSMA, 2022; Gartner, 2022). Accessible across various devices, from smartphones and tablets to VR headset displays, the metaverse is anticipated to transform how businesses and consumers engage. Global brands such as Adidas, Walmart, and McDonald's leverage metaverse platforms for brand engagement, virtual commerce, and immersive consumer experiences (Mogaji et al., 2023; Lee et al., 2024; Wasiq et al., 2024). Market forecasts from various reports indicate swift growth, projecting an increase from 22.79 billion USD in 2021 to 996.42 billion USD by 2030, with a compound annual growth rate (CAGR) of 39.8 % (Adhini and Prasad, 2024; Gartner, 2022). By 2026, 25 % of the global digital population is expected to spend at least 1 h in the metaverse daily for various activities such as work, entertainment, and shopping (Gartner, 2022).

Despite its growing adoption, metaverse engagement remains inconsistent due to technological factors like skepticism, usability barriers, and privacy concerns (Hilken et al., 2022; Zhou et al., 2024). While some consumers readily explore virtual spaces, others hesitate due to various concerns, including financial risks, digital fatigue, and data security (Hadi et al., 2024; Natarajan et al., 2024). This raises a critical research question: What drives and inhibits consumer adoption of the metaverse, and how do individual technology predispositions shape this decision-making process? Although businesses continue to invest in

* Corresponding author.
E-mail address: sigmasoni.fpm19@micamail.in (S. Soni).
[1] Sigma Soni, Parvinder Arora, and Dharun Kasilingam have contributed equally to the entire manuscript.

https://doi.org/10.1016/j.jretconser.2025.104294
Received 8 December 2024; Received in revised form 11 March 2025; Accepted 19 March 2025
Available online 25 March 2025
0969-6989/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

metaverse experiences, there is uncertainty regarding the determinants of adoption and sustained engagement. This study aims to bridge this knowledge gap by identifying both motivators and barriers affecting consumer adoption.

Existing research on the metaverse has primarily explored technological advancements (e.g., platform development, blockchain integration) and business applications (e.g., marketing, gaming, and virtual commerce) (Payal et al., 2024; Hollensen et al., 2022; Dwivedi et al., 2022). However, there is a limited understanding of the psychological and behavioral factors driving or hindering adoption of the metaverse from a consumer standpoint (Hilken et al., 2022; Zhou et al., 2024). While prior studies, including the recent articles cited in Table 1, have applied technology adoption models such as the Technology Acceptance Model (TAM) (Davis, 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003), these models focus predominantly on enablers of adoption and do not fully account for adoption barriers (e.g., trust concerns, usability challenges) (Claudy et al., 2015; Yu, 2024). Behavioral Reasoning Theory (hereafter referred to as BRT) (Westaby, 2005) provides a more comprehensive perspective, as it incorporates both "reasons for" (motivators) and "reasons against" (barriers) adoption. However, prior research has not examined technology-specific personal values within BRT, making it necessary to explore how individual predispositions influence metaverse adoption behavior. This study integrates the Technology Readiness Index (hereafter referred to as TRI) (Parasuraman, 2000) into BRT to investigate how optimism, innovativeness, insecurity, and discomfort influence consumer engagement with the metaverse (Jan et al., 2023).

This study focuses on Generation Z (1997–2012) and Millennials (1981–1996), constituting the most extensive and engaged metaverse user base. Research indicates that these cohorts are early adopters of immersive technologies like VR, AR, and blockchain, whereas older generations exhibit lower engagement due to usability concerns and privacy risks (Yu, 2024). Auxier and Arbanas (2023) study found that 29 % of Gen Z and Millennials were interested in VR-based virtual travel, compared to 21 % of Gen X and Baby Boomers. Similarly, Retail TouchPoints (2022) reported that 86 % of Gen Z and 81 % of Millennials are familiar with the metaverse, versus 68 % of Gen X and only 42 % of Baby Boomers. Additionally, this focus enables comparative analyses across income levels, gender, metaverse usage intensity, and education. Since older consumers engage less frequently, targeting Gen Z and Millennials ensures more relevant insights into the motivations and barriers influencing early adoption.

This study, hence, aims to develop an integrated framework to understand consumer metaverse adoption by incorporating psychological, behavioral, and technology-readiness factors for the Generation Z and Millennials Metaverse audience. The study addresses the following questions:

RQ1: What factors drive consumer decisions for and against adopting the metaverse?
RQ2: How does technology readiness shape metaverse adoption determinants by influencing consumer attitudes?
RQ3: How do consumer attitudes toward the metaverse impact adoption intention and engagement levels?

This study makes three key contributions. First, it extends technology adoption literature by analyzing motivators (e.g., convenience, trend affinity) and inhibitors (e.g., privacy concerns, financial risk) relevant to metaverse adoption. Second, it introduces technology readiness (TRI) into the BRT framework, offering a novel approach to examining consumer attitudes toward immersive digital environments. Finally, it explores engagement as a behavioral outcome, demonstrating how metaverse attitudes shape long-term user interaction beyond initial adoption. By addressing these gaps, this study provides practical insights for businesses, policymakers, and marketers, helping them develop strategies that encourage adoption while mitigating consumer resistance.

The rest of this paper is structured as follows. Section 2 presents the literature review, highlighting research gaps and justifying the BRT-TRI integration. Section 3 details the qualitative phase, where insights from 48 industry experts inform hypothesis development. Section 4 outlines the quantitative research methodology, including data collection from 771 Gen Z and Millennial consumers. Section 5 presents the findings and theoretical and managerial implications, while Section 6 concludes with key contributions, strategic recommendations, and future research directions.

2. Review of literature

2.1. Metaverse and its application

The Metaverse is a relatively new concept with various definitions developed in the last year. The Metaverse is the combination of 'meta,' meaning beyond, and 'verse,' from the universe. The Metaverse is "a virtual copy of the real world" (Tlili et al., 2022). According to Lee et al. (2024), the Metaverse is a domain where digital and physical worlds intersect and combine to create new social, economic, and cultural value for users. Dwivedi et al. (2022) have also explained it as the latest internet generation, an entirely new virtual world where people can interact with technology. Though the term 'Metaverse' was first coined by Neal Stephenson in his 1992 novel Snow Crash, it has since undergone a massive change in the past 30 years. According to Koohang et al. (2023), it is now poised to generate business opportunities in trillions of dollars. In the current world, consumers and companies are aware of Metaverse's significant potential as a dynamic creator economy (Accenture, 2023). As defined by Kye et al. (2021), the Metaverse is a 3D virtual reality (VR) world where people live and work, performing their daily activities and economic transactions using avatars that mimic their authentic selves (Kim et al., 2023).

This study admits that the term 'Metaverse' encompasses different virtual platforms distinct in nature and coverage. It is dynamic and evolving, but the metaverse has not yet attained its full potential (Dwivedi et al., 2024; Ooi et al., 2023). In our research, metaverse means the current platforms and technologies that are the foundation of the future infinite digital universe. As the Metaverse is still being created, progress towards its final form is necessary. We pay attention to consumer adoption and engagement with the Metaverse because we are a part of the ongoing evolution that is leading to the realization of this immense vision (Mogaji et al., 2023; Nunkoo et al., 2024).

2.2. Behavioral Reasoning Theory (BRT)

As Westaby (2005) stated, BRT is a psychological model that helps explain specific behaviors by examining the cognitive processes behind them (Claudy and Peterson, 2014). Based on earlier models, including the Theory of Reasoned Action and the Theory of Planned Behavior, BRT has unique advantages over other behavioral theories (Ryan and Casidy, 2018; Westaby, 2005). It includes both the supportive and the opposing reasons, thus providing a more detailed perspective on the decision-making process and valuable information regarding users' intentions. For example, supporting 'Reasons for' assists in adopting new technologies, while opposing, for example, reasons against,' represents barriers.

BRT is concerned with how values and beliefs affect decision-making and implementation (Sahu et al., 2020; Westaby, 2005). Research shows that BRT models better explain the variation in dependent variables than other behavioral theories (Claudy et al., 2015). As core values, principles help people determine whether certain behaviors are consistent with their standards and concerns. Past studies have shown the efficiency of BRT in explaining consumer behavior in complex choices, including the selection of organic foods (Tandon et al., 2020), the adoption of innovations (Claudy et al., 2013, 2015), and the uptake of

Table 1
Recent studies examine various perspectives on the metaverse.

Source	Study Context	Theory	Methodology	Factors	Key Findings
Ghali et al. (2024)	Tours and Tourism Service Marketing	Stimulus Organism Response	Experimental	Amount of time spent in metaverse, Number of friends in metaverse, Use of virtual places, Social presence in metaverse, Attachment to metaverse, Engagement to metaverse, (Re) visit intentions	The research found that time spent, social connections, and virtual space use in the metaverse boost Generation Z's social presence and attachment, which increases their intention to revisit. Social presence and attachment mediate these effects, with engagement enhancing their impact on return intentions.
Alexander et al. (2024)	Fashion Industry	Customer Experience Framework	Qualitative	Freedom of expression, Creativity, Exploration, Entertainment, Productivity, and Customer experience	The findings suggest that out of the five motivations to use the metaverse; the environmental dimension significantly influences the customer experiences in the metaverse- a key driver for creating immersive and interactive experiences.
Bilgihan et al. (2024)	Metaverse Brand	Metaverse Engagement Model	Theoretical analysis, Practical case studies	NA	The study reveals that personalized experiences in the metaverse engagement model increase brand visibility and customer loyalty.
Ki et al. (2024)	Digital Fashion	NA	Qualitative	Digital fashion adoption, Transitory metaverse engagement, Avatar personalization, and branded digital fashion	The findings highlight three primary consumer motivations in the evolving metaverse: a peaceful escape, virtual ownership and empowerment, and transitional interaction. Avatar customization is driven by three key themes: self-expression based on mood, freedom from societal expectations, and social recognition. Users viewed their avatars as virtual doubles, idealized versions of themselves, or digital figures. Four themes emerged regarding branded digital fashion adoption: visual appeal, affordability, connection to the real world, and alignment with favorite real-world brands.
Gupta et al. (2024)	Service Industry	Force field theory, Technological, organizational, and environmental (TOE) framework	Fuzzy analytic hierarchy process, Fuzzy decision-making trial and evaluation laboratory	Security and Privacy, Governance and Standardization, Infrastructure, User Behavior and psychology, Feasibility, and Organizational culture and Stakeholder commitment	The study finds that Security and Privacy are the most significant barriers to metaverse adoption in the service sector, followed by Governance and Standardization and other factors like Infrastructure and User Behavior. Using FAHP and F-DEMATEL, the study identifies the connections between these barriers, offering insights into their impact and a framework for addressing them.
Chakraborty et al. (2023)	Dating in metaverse	Elaboration Likelihood Model	Mixed method: Python and Confirmatory analysis	Perceived accuracy of the information, Perceived interaction quality, Avatar customization, Perceived visual attractiveness, Perceived intrinsic enjoyment, Perceived entertainment value, Perceived attitude, Realistic threat, Perceived trust, Dating intention in the metaverse	The study reveals that perceived trust influences the link between perceived attitude and dating intentions in the metaverse, whereas realistic threat does not. Most factors notably affect perceived attitude, except for the perceived accuracy of information. It's crucial to address privacy and security concerns through strong user verification and security measures to build user trust in metaverse dating.
Mehrotra et al. (2024)	Metaverse Retailing	Uses and Gratification Theory 2.0	Structural Topic Modeling	Modality, Agency, Interactivity, Navigability, Customer Experience	The study explores four key themes in metaverse retailing—immersive technology, controlled interaction, agency interface integration, and novel environments—using Uses and Gratification Theory 2.0 (UGT 2.0). It proposes three research models to enhance customer experience by integrating UGT 2.0 with Flow Theory, Cognitive Load Theory, and Presence Theory, advancing metaverse retailing research.
Kumar et al. (2024)	Enterprise Metaverse and Employee Perspective	Diffusion of Innovation	Quantitative (AMOS 26)	Relative advantage, Compatibility, Complexity, Observability, Triability, Digital agility, Intention to use enterprise metaverse, Government policy effectiveness	The study finds that Diffusion of Innovation (DOI) theory dimensions significantly influence enterprise metaverse adoption intentions, with digital agility mediating this relationship. The mediating effect of digital agility is further moderated by government policy effectiveness, providing valuable insights

(*continued on next page*)

Table 1 (*continued*)

Source	Study Context	Theory	Methodology	Factors	Key Findings
					into employee adoption behaviors in enterprise metaverse settings.
Taylor et al. (2024)	Avatar Creation	Self-presentation & Identity formation	Qualitative (In-depth semi-structured interviews)	Consolidation, Exploration, Affirmation, and Aspiration.	This mixed-methods study examines metaverse adoption, exploring drivers and barriers through qualitative and quantitative approaches. Grounded in BRT, it focuses on adoption intentions and customer engagement, addressing gaps in existing literature.
Jafar et al. (2023)	Metaverse stores	Stimulus Organism Response	Quantitative (PLS-SEM)	Effectiveness, Vividness, Interactivity, Metaverse telepresence, Perceived product knowledge, and Purchase intentions in the metaverse	The study shows that key functional attributes of the metaverse—effectiveness, vividness, and interactivity—boost telepresence, which positively impacts product knowledge and purchase intentions in metaverse stores. These enhanced technological experiences significantly shape consumer behavior, supporting the S–O–R framework and aligning with previous studies on virtual retailing.
Akour et al. (2022)	Metaverse for the Higher institutions of the Gulf area	Technology Adoption Model	Quantitative (Deep-learning-based analysis of structural equation modeling and artificial neural network-ANN, followed by Importance performance map analysis)	Adoption properties such as Perceived trialability, observability, User compatibility, complexity, Personal innovativeness, and TAM constructs such as Users' satisfaction, perceived ease of use, Perceived usefulness, User's intention to use metaverse	The study identifies Perceived Usefulness (PU) as the most significant predictor of students' intention to use the metaverse for education, supported by ANN and IPMA analyses. Additionally, it highlights the effectiveness of deep-learning-based ANN architecture in uncovering non-linear relationships within theoretical models, aiding strategic planning in education.
Xi et al. (2024)	Retail Metaverse Shopping	Technology Acceptance Model (TAM)	Experimental	AR, VR, Perceived enjoyment, Perceived ease of use, Perceived usefulness, Attitude, Intention to visit	The study found that AR hurts perceived ease of use, while VR improves perceived enjoyment in metaverse shopping. Both perceived enjoyment and usefulness strongly influence consumer attitudes and intentions to revisit, with ease of use surprisingly showing no adverse effect on adoption.
Adhini and Prasad (2024)	Metaverse	Technology Acceptance Model (TAM), Unified Theory of Acceptance and Use of Technology (UTAUT)	Mixed Method: Qualitative: In-depth semi-structured interview (n = 20), and Quantitative: PLS-SEM (n = 1250)	Perceived enjoyment, Perceived ease of use, Perceived usefulness, Attitude, Behavioral intention, Personal innovativeness, Facilitating conditions, Telepresence, Social presence, Trust, Regulatory support	The study's findings show that perceived usefulness, personal innovativeness, perceived enjoyment, facilitating conditions, telepresence, social presence, trust, and regulatory support impact attitudes and behavioral intentions toward the metaverse. Additionally, the study highlights significant variations in the strength of certain relationships across age, gender, and income levels, offering valuable insights for organizations aiming to target different demographic groups strategically.
Yu et al. (2024)	Metaverse Retail	Technology Acceptance Model (TAM),	Quantitative (n = 329), Confirmatory factor analysis AMOS 26	Usefulness, Ease of use, Sensory experience, Behavioral experience, Social presence	The study found that usefulness and ease of use positively impact sensory experience, with ease of use also enhancing behavioral experiences. The findings highlight varying levels of influence usefulness and ease of use have on user experiences within metaverse contexts. Additionally, the results clarify the mediating effect of sensory experience and the moderating role of the sense of presence.
Natarajan et al. (2024)	Metaverse	Uses & Gratification Theory	Mixed Method: Qualitative: Structural topic modeling to analyze the tweets, Quantitative: SEM-PLS	Convenient navigability, Escapism, Social interaction, Telepresence, Trendiness, Intention to use metaverse, Recommendation to use metaverse	The study identified five key gratifications influencing metaverse adoption: escapism, social interaction, convenient navigability, telepresence, and trendiness. Escapism and social interaction emerged as the most significant, with escapism driving stress-free, immersive experiences and social interaction fostering meaningful connections, particularly among female users. Convenient navigability, valued more by Gen Y, enhances seamless transitions and user satisfaction, while telepresence strengthens realism and engagement.

(*continued on next page*)

Table 1 (*continued*)

Source	Study Context	Theory	Methodology	Factors	Key Findings
					Trendiness had minimal impact, as users prioritized functionality and value over novelty.
Ahn et al. (2024)	Metaverse	Self-expansion theory	Quantitative: AMOS 28 & SEM	Perceived interactivity, Gamification, Control, Playfulness, Synchronicity, Social engagement, Connectedness, Responsiveness, Self-expansion, Purchase intention of virtual items, Metaverse access frequency	The study discovered that connectedness and playfulness in the metaverse foster self-expansion, boosting the intention to purchase virtual items as users perceive them as part of their identity. However, control, synchronicity, and responsiveness did not significantly impact the situation. Additionally, frequent use of the metaverse diminished the influence of gamification and social engagement on purchase intentions due to hedonic adaptation, where the excitement and enjoyment of experiences diminish over time.

mobile banking services (Gupta and Arora, 2017). Applying BRT to the Metaverse is vital, considering the latter's innovative and immersive characteristics, which require a more detailed examination of the psychological factors that may affect users' adoption. The Metaverse is a compelling shift in digital engagement that appeals to users to consider various benefits (including virtual presence and improved connectivity) against multiple challenges (including technological difficulties and skepticism), for which the dual reasoning framework of BRT is a suitable theoretical perspective.

Furthermore, while values explored in prior BRT studies often focus on general principles like universalism or openness to change, these may not fully capture the technology-specific considerations critical to metaverse adoption. To address this, the study incorporates the TRI as a construct that bridges broader value systems with technology-specific predispositions.

2.3. Technological readiness

Technology readiness (TR) is a psychological term that describes how people's minds react to utilizing and integrating new technologies into their daily lives. In particular, they defined TR as people's propensity to embrace and use the latest technologies to achieve specific goals in both domestic and working life (Parasuraman, 2000, p.308). Researchers have pointed out that technology readiness is a stable individual difference that is not easily changed or manipulated (Parasuraman and Colby, 2015; Cruz-Cárdenas et al., 2021).

Parasuraman (2000) conceptualized technology readiness as a four-component multidimensional framework: Innovativeness, Optimism, Discomfort, and Insecurity, and he provides a balanced view by recognizing the motivators (Optimism, Innovativeness) and inhibitors (Discomfort, Insecurity) of technology adoption. Together, these elements determine how people perceive and use technology. The dual perspective of the TRI model makes it suitable for examining metaverse adoption, which is a domain that is both rewarding (e.g., through immersive experiences and digital freedom) and challenging (e.g., usability and privacy issues) (Cruz-Cárdenas et al., 2021; Parasuraman and Colby, 2015).

In the metaverse context, those high on optimism and low on innovativeness may have a favorable view of the platform as a creative and forward-thinking innovation consistent with their core values of optimism and progress. On the flip side, those who are more likely to exhibit discomfort or insecurity in the context of the metaverse tutorial may perceive the metaverse as more complicated and risky, which will result in more reasons for people not to adopt the metaverse (Jan et al., 2023; Parasuraman and Grewal, 2000). Integrating TRI's motivators and inhibitors with BRT's reasoning constructs is thus seamless, providing a rich understanding of why people adopt or reject the metaverse.

As opposed to other models, such as the Technology Acceptance Model (Davis, 1989) or the Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003), which are primarily concerned with perceptions of usefulness and ease of use, TRI provides a more value-laden and personality-based perspective. This is important in the Metaverse, where user engagement depends on functionality and users' beliefs about technological progress, security, and personal creativity. Whereas TAM or UTAUT are typically used to predict rational, functional decisions, BRT and its implementing constructs (such as TRI) also incorporate logical reasons (e.g., ease of use) and emotional or value-driven considerations (e.g., optimism or discomfort).

Thus, this study has holistically addressed the complexity of metaverse adoption by integrating TRI with BRT to understand how predispositions toward technology affect both 'reasons for' and 'reasons against' usage. This contributes to theoretical advancement and practical implications for marketers and developers to develop specific strategies consistent with users' values and readiness. Furthermore, as Table 1 illustrates, researchers have considered several metaverse adoption factors to assess the advantages of customer familiarity with the store, shopping expertise, and user privacy and security concerns. This analysis applies different theories, including the "stimulus, organism, and response theory," the "uses and gratification theory," and the "technology acceptance model" across diverse fields.

Table 1 shows that metaverse applications can become disruptive in the fashion, retail, tourism, services, healthcare, and hospitality industries. Therefore, it is crucial to understand the 'reasons for' and 'reasons against' metaverse adoption from the consumer's perspective to enhance its application across various domains. This study aims to provide in-depth insights into consumers' motivations and concerns regarding metaverse adoption.

3. Research methodology: *mixed methods approach*

This study uses a mixed-methods research design with both qualitative and quantitative approaches to explore the factors influencing metaverse adoption, focusing on both the drivers and barriers, an area that remains underexplored in previous literature (Adhini and Prasad, 2024; Yu, 2024). The key purpose of this approach was to understand adoption intentions and customer engagement with BRT as the theoretical framework focusing both on the divers and deterrents. The choice of a mixed-methods approach is advantageous, as it leverages both qualitative and quantitative research methods to provide a more comprehensive and robust understanding of the phenomenon being studied (Venkatesh et al., 2016), especially in the context of an emerging technology like the metaverse (Natarajan et al., 2024; Payal et al., 2024). The qualitative phase of the research was used to identify key themes and vital insights acquired from respondents' experiences with

metaverse adoption. The quantitative phase enabled empirical testing of the relationships and constructs that emerged from the qualitative findings (Creswell and Clark, 2017). Fig. 1 explains the study's research design. In the first phase, the qualitative study was conducted by interviewing industry experts (n = 48) to uncover the 'reasons for' and 'reasons against' Metaverse adoption. These qualitative findings were then used to create a theoretical model in the second stage, which was tested using survey data gathered from 771 customers. By embracing qualitative and quantitative methodologies, this study aims to create a more nuanced understanding of the complex dynamics influencing metaverse adoption, ultimately contributing novel insights to the field.

3.1. Qualitative-inductive research design

Consumer adoption and engagement with the metaverse is nascent, with limited academic knowledge and relevant theoretical frameworks (Mogaji et al., 2023). Hence, a qualitative inductive research design was considered to gain initial and exploratory insights (Yu, 2024; Zallio and Clarkson, 2022). The study used a constructivist and interpretivist paradigm to explore the 'reasons for' and 'reasons against' adopting the metaverse. It uses semi-structured interviews that give respondents in-depth insights and viewpoints on the metaverse (Kaur et al., 2024). This methodology for the data collection is aligned with past investigations, where researchers have used comprehensive interviews as an essential tool for qualitative research in many technology-oriented management and consumer behavior studies (Natarajan et al., 2024; Adhini and Prasad, 2024; Mogaji et al., 2021).

3.1.1. Sample participants and data collection

Our qualitative study focuses on early adopters, specifically tech experts and managers who have seamlessly integrated the metaverse into their business practices. Data collection was facilitated through judgmental and snowball sampling methods (Parker et al., 2019). We targeted technology specialists and managerial professionals engaged in various aspects of tech innovation and marketing related to the metaverse. We selected interviews based on two criteria. They (1) Have already ventured into the metaverse with their brand or are actively considering doing so. (2) It specifically included individuals holding top- or mid-level managerial positions and technopreneurs actively leveraging metaverse technologies for their business initiatives.

Interviewing industry experts on metaverse adoption intentions offers unique insights beyond consumer research (Eggenschwiler et al., 2024). Accordingly, we conducted semi-structured interviews with 48 experts using a discussion guide (Refer to Appendix 1) (Gaskell, 2000). Participants received detailed research context instructions before online interviews, ensuring a comfortable, judgment-free environment that encouraged diverse perspectives.

Marshall and Hubbermas recommend a sample size of 20–50 for valid results (Van Esch et al., 2019). The respondents' demographics are detailed in Appendix 2, while Table 2 presents the demographic profile analysis for the qualitative study. Among 48 professionals, 25 % had a core tech background, 52 % were digital innovators, and 23 % were UX/UI developers and brand strategists. The interviews followed a structured guide based on a literature review and BRT principles (Engel et al., 2001). Two management science professors validated the guide, followed by a pilot study with eight industry professionals (three females, five males) to ensure effectiveness.

Table 2
Analysis of respondents' profile.

Age Range 23–53 (in numbers & percentage)		Gender		Designations	
18–25	9 (19 %)	Male	31 (65 %)	Technopreneurs	25 (52 %)
26–33	15 (31 %)	Female	17 (35 %)	Digital innovator	25 (12 %)
34–41	14 (29 %)			UX/UI developers and brand strategists	23 (11 %)
42–49	8 (17 %)				
>50	2 (4 %)				

3.1.2. Interview structure

The pilot interviews included four technopreneurs from metaverse development companies and four specialists in leadership roles at well-known IT companies. After the pilot study, interviewees were chosen based on three different criteria. (1) sufficient understanding of the metaverse, virtual world, and related technologies; (2) if they have previously worked in pilot projects involving metaverse technologies for brand-customer engagement and experience with them; and (3) intentions to adopt metaverse technologies in the future for their brands. The interviews were conducted via video conferencing, recorded, and transcribed.

3.1.3. Data analysis

The interview duration ranged from 30 min to 1.5 h, with an average time of 50 min. The interviewer collected information on attitudes, behaviors, experiences, views, and arguments for and against adopting the metaverse. The first forty-four interviews were sufficient to reach theoretical saturation. As a result, the 48-individual sample was adequate to get the results. The standard methods Cresswell (2013) recommended were followed in examining the qualitative data. All interviews were recorded and subsequently transcribed verbatim. A line-by-line content analysis was conducted on the transcripts, following the recommended procedures for qualitative research (Miles, 1994). Keywords or phrases that provided the context of reasons for and reasons against metaverse adoption were sought. To analyze the data for this study, the authors became familiar with and immersed themselves in the data, including reading interview transcripts, following the six steps of thematic data analysis described by Braun and Clarke (2006). The data was then categorized using an inductive approach, and each category was labeled.

3.1.4. Coding

Coding required continuous iteration between theory and data as codes were gradually integrated and modified in response to the available data. This development of the codes was based on emerging information and existing related research. The subsequent phase uncovered the crucial themes (reason for and reason against) that

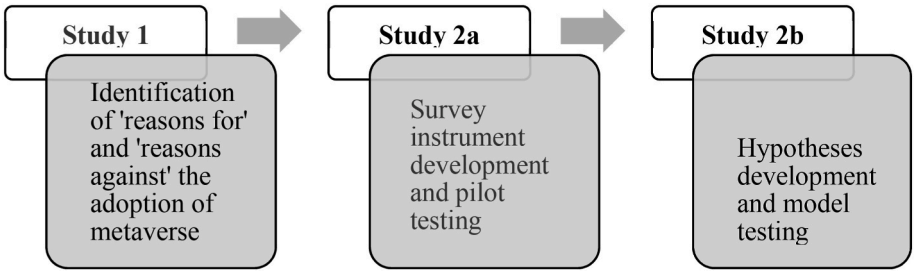


Fig. 1. Sequential research design of the current study.

resulted from the investigation and generated detailed descriptions of each theme (refer to Table 3).

One research scholar and two professors of management science developed the codes. We used a three-phase approach to evaluate the dependability of our coding procedure. First, we combined the independent coding of each researcher to ensure internal consistency. To ensure inter-coder reliability, three academic experts in qualitative research approaches who were unfamiliar with our study were then brought in to review the interview data coding data using the open, axial, and selective coding processes (Williams and Moser, 2019). The inter-coder reliability among the three coders was 0.92. The value is much above the acceptable threshold. Additionally, we verified the idea by asking interviewees for their opinions on the defined categories.

3.2. Findings of a qualitative study

Table 3 represents the 'reasons for' and 'reasons against' adopting the metaverse. The second-level categories' four concepts for 'reason for' are 1) Haptic imagery, 2) Convenience, 3) Trend affinity, and 4) Relative advantage. Further 'reasons against' those observed in 2nd-level categories are 1) Privacy concern, 2) Risk of malfunction, 3) Financial concern, and 4) Perceived performance risk profoundly impacts the adoption of metaverse services (see Table 3 for details).

3.2.1. Haptic imagery

Haptic Imagery incorporates sensory feedback, predominantly touch-related cues, enabling users to virtually experience a sense of touch (Ivanov et al., 2023). As users encounter a more vivid and realistic representation of products, they are likely to develop a stronger sense of engagement and connection within the social aspects of the immersive platform, like AR/VR (Trivedi et al., 2022; Shen et al., 2023). Users interacting with products through tactile simulation fosters a deeper sensory engagement, leading to a more authentic and immersive experience (Brakus et al., 2009; Shen et al., 2023). Erensoy et al. (2024) found that consumers actively seek and respond positively to haptic stimulation online shopping through mediums like Metaverse. For example, the respondent mentioned *(P13), "I love the idea of tactile feedback in the virtual world. It makes everything feel so much more real and immersive for me." (P37), "Being able to feel textures and surfaces through tactile sensations makes the experience way more interactive and convincing for me."*

3.2.2. Trend affinity

Trend affinity arises when consumers follow a trend or desire to engage with high-tech products and services, such as the Metaverse (Möhlmann, 2015). Numerous studies reveal that consumers value new technologies like robot chefs and AR/VR for their efficiency and hedonic benefits, ultimately boosting enjoyment and satisfaction. Notably, it has been found that trend affinity positively influences customers' behavior toward new technology (McLean and Osei-Frimpong, 2019). For instance, recent research demonstrates that aspects of self-identity, such as viewing oneself as tech-savvy (where technology is considered a marker of trendiness), can effectively predict behaviors related to adopting new technologies like Metaverse (Jung et al., 2023; Kautish et al., 2023). One respondent stated *(P20), "I like staying up to date with the latest trends, and the metaverse feels like the next big thing I do not want to miss out on." (P11). Everyone is talking about the metaverse, and we and our brand want to be part of this new trend. It feels exciting to be on the cutting edge to offer unique experiences to our customers".*

3.2.3. Convenience

Convenience was described by Copeland (1923) as the amount of time and effort a customer spends using a service or making a purchase. This concept is a crucial component of the TAM, which emphasizes perceived convenience and its role in facilitating the ease of task completion (Iba'nez-Sánchez et al., 2022). In today's fast-paced world,

Table 3
Perspectives and emerging themes/thematic analysis of the reasons for and reasons against using the metaverse.

No	Open code	Axial code	First order code	Interview quotes
Reason for the "Metaverse Adoption"				
1	Physical interaction enhancement	Strengthening customer-brand relationships through immersive experiences	Haptic Imagery	*"Integrating haptic imagery in the metaverse allows our consumers to feel our products physically, enhancing their virtual shopping experience and making them more likely to purchase."*
	Deeper customer connection			*"We thrive on creating a more immersive brand experience through metaverse, enabling customers to connect with our brand on a deeper level."*
	Unique interactive experiences			*"By adopting this technology, we're not just selling products; we're offering a unique experience you can feel, differentiating our brand in a crowded digital marketplace."*
2	Operational efficiency	Market reach and speed	Convenience	*"The metaverse allows us to be more efficient than our competition simply can't match. It's about speed and efficiency for us."*
	Time reduction			*"Metaverse has helped us significantly reduce the time spent marketing our products and enable us to reach the global level at our fingertips."*
	Flexibility in reach			*"The convenience of virtual showrooms in the metaverse significantly reduces the time and resources needed for physical exhibitions and engage with customers."*
3	Technological innovator	Brand positioning and innovation	Trend Affinity	*"Embracing the metaverse allows us to tap into the growing enthusiasm and affinity for immersive digital experiences, particularly among younger consumers shaping future trends."*
	Staying ahead of the curve			*"For us as a brand, staying ahead of the curve is a must. Our*

(continued on next page)

Table 3 (*continued*)

No	Open code	Axial code	First order code	Interview quotes
				involvement with the metaverse signals to our competitors and consumers alike that we are leaders in embracing new digital realms."
	Trend alignment for users			*"We decided to join the metaverse because more and more of our target audience interacts with brands online. We need to be where they are."*
4	Unmatched brand experience	Adopting metaverse to gain competitive advantage	Relative Advantage	*"The metaverse presents a distinctive platform enabling the creation of immersive and interactive brand experiences, unparalleled by any other digital or physical channels. It centers around delivering unparalleled value to our consumers."*
	Competitive advantage			*"Metaverse allows us as a brand to shape our virtual landscape and establish a leadership position before the space becomes saturated."*
	Global Engagement			*"Incorporating elements of the metaverse into our marketing approach has given us a distinct competitive edge. It enables us to engage with our audience through groundbreaking methods and offer exceptional value that distinguishes our brand."*
Reason Against the "Metaverse Adoption"				
5	Data Risks	Maintaining data security and ethical concerns	Privacy Concerns	*"The constant tension of knowing that the metaverse gathers a substantial amount of personal data might raise significant problems. We are hesitant to expose our brand and customers to potential data breaches."*
	Customer Distrust			*"Our customers value their privacy, and adopting the metaverse might raise ethical concerns about data usage and personal information."*
	Brand reputation			*"Privacy issues in the metaverse could damage our brand reputation if mishandled or breached."*
6	Technical glitches	Safeguarding Against Security Vulnerabilities	Risk of Malfunction	*"The tension is constant since we're always worried about tech problems and system errors in the metaverse messing up how we connect with customers and how they see our brand."*
	Infrastructure readiness			*"Sometimes I feel that we do not have sufficient infrastructure to deal with operational glitches in the metaverse, which is also responsible for the financial losses."*
	Security malfunction			*"The possibility of hardware or software malfunctions in the metaverse introduces concerns about dependability, which could adversely affect our brand's image."*
7	High Initial cost	Financial risk assessments	Financial Concerns	*"Integrating our brand into the metaverse requires a substantial upfront investment, and with the current unpredictability of the digital market, it's hard to justify the cost."*
	Uncertainty on returns			*"Looking at current market conditions, we are hesitant to allocate resources to the metaverse without clear ROI metrics and a solid business case."*
	Recurring cost concerns			*"The ongoing expenses related to maintaining a presence in the metaverse, such as digital asset creation and platform fees, could strain our budget, which is already allocated to proven strategies."*
8	Technical reliability	Associated performance related risk while using the new technology	Perceived Performance Risk	*"One of the key reasons that we are hesitant to integrate metaverse completely in our business is that we*

(*continued on next page*)

Table 3 (*continued*)

No	Open code	Axial code	First order code	Interview quotes
				are concerned about the potential performance issues in the metaverse, such as lagging or latency problems during virtual events."
	Disconnect with brand			*"Introducing our brand into the metaverse might not meet our customers' expectations, leading to a disconnection between our digital and physical branding efforts."*
	Doubts regarding performance			*"The technology behind the metaverse is still evolving. If it doesn't perform reliably, it could negatively impact the user experience and, by extension, our brand's reputation."*
9	Immersive experiences	Enhancing customer bond and interaction	Brand Engagement	*"Engaging in the metaverse allows us to create immersive experiences that connect with our audience on a deeper level than traditional digital platforms."*
	Real time interaction			*"The metaverse space provides unique opportunities for real-time interaction with our brand, making our engagements more dynamic and memorable."*
	Interactive experiences			*"By utilizing the metaverse, we're not just selling products; we're providing a platform for our customers to experience our brand's world, which significantly boosts engagement."*

where time is a finite resource, convenience is primarily associated with time savings (Yale and Venkatesh, 1986; Roy et al., 2020). The materialization of Metaverse made individuals realize the potential advantages like cost effectiveness, sustainability, and convenience (Buhalis et al., 2023). Respondents in our study also reciprocated this thought well; for instance, one participant mentioned, *"Metaverse is so convenient. I can make consumers attend events and even shop without leaving home" (P47)*. Another respondent shared, *"The Metaverse offers great convenience. My brand can interact with people, make consumers explore new offerings, and let them complete tasks without the hassle of real-world logistics" (P33)*. Based on the above rationale, the authors recommend that a favorable assessment of the Metaverse as a practical and time-saving tool will lead to higher engagement at both the pre-use and usage phases.

3.2.4. Relative advantage

It refers to "the degree to which an innovation is perceived as being better than the idea it supersedes" (Rogers, 2003, p. 229). Customers' propensity to accept innovation might be positively impacted by their perception that a new service is superior to more traditional ones. Metaverse adoption may outpace conventional industry practices, enabling more efficient virtual operations and overcoming physical constraints (Kaur et al., 2024). As one respondent stated *(P32), "I feel that the metaverse allows me, as a brand, to shape my virtual landscape and establish a leadership position before the space becomes saturated." (P27). "I believe the metaverse presents a distinctive virtual platform that enables me to create immersive and interactive brand experiences unparalleled by any other digital or physical channels. It allows me to focus on delivering unmatched virtual value to my consumers".*

3.2.5. Privacy concern

According to the literature, one of the drawbacks of new technology is privacy concerns. These might include worries about collecting and storing personal data, using data without authorization, misusing private information, and unapproved sharing with third parties (Choi et al., 2018; Xu et al., 2013). They can also include worries about data collection, data mining, and information retrieval (Vimalkumar et al., 2021). In mainstream news, the metaverse presents fundamental privacy concerns and challenges (Zhao et al., 2023a,b). Mainly in the context of metaverse, Dwivedi et al. (2022) found that users are not keen on using metaverse technology because of privacy concerns. Additionally, when users interact with others to exchange information or participate in different marketing activities within the metaverse, privacy problems arise, including personal information leakage. As one respondent stated *(P7), "When I think of using certain technology, I am more concerned about how my personal information will be used and kept intact."* (P21) *"For me, using a new technology requires security. Because these things become very risky otherwise."*

3.2.6. Perceived performance risk

The main focus of the research reported here is perceived performance risk, which Stone and Grønhaug (1993) defined as the risk linked with the failure of Metaverse technology. Consistent with Sweeney et al. (1999, p. 81), perceived risk is defined here as "the subjective expectation of a loss" (Sweeney et al., 1999, p. 81). Performance risk is one of the characteristics of perceived risk established by previous studies (Lowe, 2010; Sun et al., 2010). Customers and potential users may quickly identify performance risks because any early-stage product or service is not yet technically mature or well-known. Combining data from various I.P. infrastructures for steady operation presents technical problems for global metaverse services (Yaqoob et al., 2023). Users may start to worry about the Metaverse's functionality if communication inside it is hindered. As one respondent said *(P46), "I am hesitant about the metaverse because I am not convinced it will perform as promised. The idea of investing time and resources into something that might not meet expectations is a significant barrier for me." (P19) "I am unsure about diving into the metaverse because I am uncertain about its reliability. If there are performance setbacks, it could impact the experience and overall perception."*

3.2.7. Risk of malfunction

The risk of malfunction is the fear of technology failure, which might result in reduced operational capacity, downtime, or data loss (Man et al., 2020; Waung et al., 2021). This fear stems from its capacity to impact the user experience since it is a key determinant of technology adoption and engagement. It has been observed that new technology makes users hesitant to participate. The reason for hesitation is fear of system failures, glitches, and software performance issues (Zhang et al., 2024). These concerns for the metaverse are inevitable, as it is new for

users (Dwivedi et al., 2022). Users of Metaverse fear performance instability, such as lag or system crashes, which could disrupt their experience and immersion, posing a significant barrier to widespread adoption (Dwivedi et al., 2022). Past studies also note that when users perceive technology as unreliable, it negatively impacts trust, engagement, and adoption (Featherman et al., 2021). As one respondent noted (P30), *"I feel we do not have sufficient infrastructure to deal with operational failures in the metaverse, which lead to financial losses."* Another respondent noted (P5*), "It is not just about the risk of potential glitches. The broader implications of those malfunctions are customer experience and trust in our brand. That is one reason our brand is holding back from the investment in Metaverse."*

3.2.8. Financial concerns

The fourth barrier pertains to the financial implications of adopting new technology. Many individuals and organizations face various financial concerns, including the costs associated with devices or services, potential installation expenses, maintenance, repairs, and the feasibility of integrating the new technology (Featherman et al., 2021). Numerous studies indicate that the financial burden of initial costs can hinder the adoption of new technologies like the metaverse. (P39) "I want to explore the metaverse, but the financial burden of buying all the gear is holding me back. It just seems too much right now." Common fears and biases against such technologies often include safety, privacy, and financial risks (PwC Report, 2022). As one respondent stated *(P9), "I am worried about the financial investment. Getting into the metaverse seems like it could be pricey, and I am not ready for that kind of commitment." (P24). "The cost of the equipment and maintaining it feels like a big hurdle. I do not want to spend much money to access the metaverse."*

4. Proposed theoretical framework and hypothesis development

We created the study's theoretical model using data from the body of existing literature as well as the results of the qualitative phase (Fig. 2). We utilized the BRT postulates to classify the outcomes into 'reasons for' (Haptic Imagery, Relative Advantage, Convenience, and Trend Affinity) and Reason Against (Risk of Malfunction, Privacy Concerns, Perceived Performance Risk and, and Financial Concerns) adopting the Metaverse. The second-order factors in the suggested model are reasons for and against, each composed of four 1st -order factors (Marsh and Hocevar, 1985). When the first-order elements are correlated, a 2nd -order model makes sense (Claudy et al., 2015). The necessity of a second-order factor model is supported by the link between the 1st -order factors, or "reasons for" and "reasons against" (Chen et al., 2005).

4.1. Reasons for adopting metaverse

The metaverse attracts consumers through compelling factors influencing their willingness to adopt this innovative technology. Grounded in BRT, "reasons for" adoption are positive drivers that shape consumer attitudes and behaviors. Based on qualitative findings and extant literature, four critical enablers have been identified: haptic imagery, convenience, trend affinity, and relative advantage. These factors highlight the metaverse's ability to deliver superior experiences compared to traditional media (Opazo-Basáez et al., 2022).

4.1.1. Haptic imagery

Haptic imagery enhances consumer attitudes toward the Metaverse for real by giving tactile sensations that make the experiences more realistic and immersive (Silva et al., 2021). Using wearable technologies like haptic gloves, users can 'feel' and engage with virtual objects, closing the distance between the digital and physical worlds (Shen et al., 2023). This multisensory feedback is essential to build trust and positive perceptions in virtual environments (Ivanov et al., 2023). In virtual retail, the ability to touch or feel how the product feels or to simulate the use of the product increases satisfaction and purchase intent (McLean and Osei-Frimpong, 2019), while in educational and professional settings, haptic feedback improves learning outcomes and collaboration,

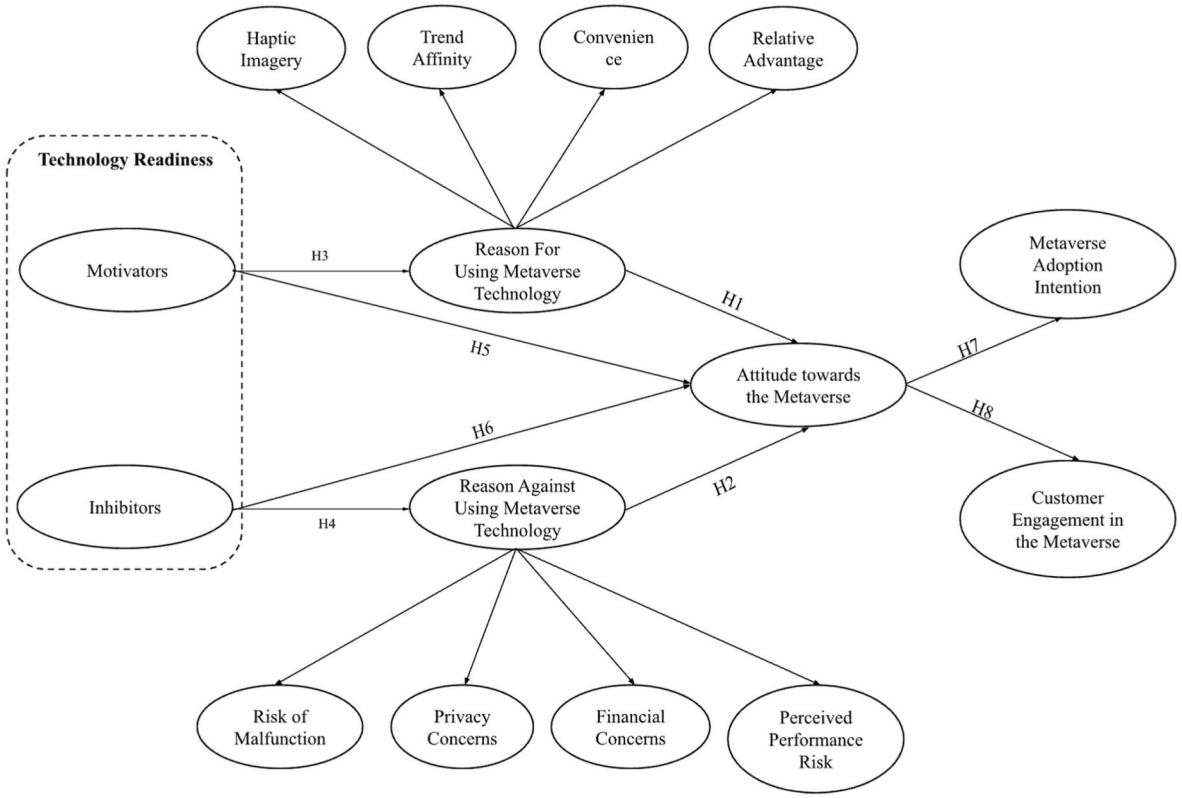


Fig. 2. The study's theoretical model (adapted from Westaby, 2005).

which in turn enhances the perceived value and usefulness of the technology (Sagnier et al., 2020). In their research, Silva et al. (2021) discussed how haptic imagery can increase attitude by integrating digital and physical interactions. A positive attitude is imperative as the investment in immersive experiences and infrastructure increases (Zhao et al., 2023a,b). Many participants in the qualitative study also highlighted the role of sensory involvement in improving experience within the Metaverse. Hence, the authors suggest that-

H1a. Haptic imagery positively influences consumer attitudes toward the metaverse

4.1.2. Convenience

Convenience, traditionally defined as the time and effort consumers expend to complete a task or use a service (Copeland, 1923), is a pivotal motivator in the virtual ecosystem (Rodríguez-Ardura et al., 2025). It enables users to perform activities such as shopping, networking, and attending events more easily (Hennig-Thurau et al., 2023; Buhalis et al., 2023; Roy et al., 2020). The metaverse's real-time interactions and ability to deliver personalized, immersive experiences align with consumer expectations for instantaneous and context-specific service delivery, enhancing its appeal in a fast-paced digital world (Payal et al., 2024). Convenience is positioned as a key factor in fostering favorable customer sentiments and increased engagement owing to these adaptable features and smooth accessibility (Eggenschwiler et al., 2024; Chekembayeva et al., 2023). A study by Gupta and Arora (2017) on adopting m-banking found that convenience influences the attitudes of individuals. Therefore, the authors propose the following hypothesis:

H1b. Perceived convenience positively influences consumer attitudes toward the metaverse.

4.1.3. Trend affinity

Trend affinity is a critical reason for a positive attitude towards the Metaverse, as it fulfils the desire to adopt and own the latest technologies (McLean and Osei-Frimpong, 2019). Past studies reveal that consumers are attracted to platforms that support their identity as tech-savvy and early adopters (Jung et al., 2023). Trend affinity is a psychological motivator that drives adoption by satisfying the need for modernity and status (Jung et al., 2023; Kautish et al., 2023). Metaverse has the potential to embody cultural trends and unique immersive experiences. The platforms incorporating recent and pop culture trends support the idea of platform usage as an expression of trendiness (Lee et al., 2024). Past studies on digital platforms also establish that trend affinity positively affects attitude towards the platform (Pantano et al., 2017; Huang and Qian, 2021). Consumers regard trend-driven adoption as expressing themselves and seeking social proof (Pantano et al., 2017; Huang and Qian, 2021). Therefore, the following hypothesis is proposed:

H1c. Trend affinity positively influences consumer attitudes toward the metaverse

4.1.4. Relative advantage

Relative advantage refers to the perceived superiority of the metaverse over traditional platforms (Erskine et al., 2019) and the extent to which an innovation is seen as better than its predecessor (Ramayah et al., 2013). The metaverse offers immersive, interactive, and scalable experiences, allowing businesses to create innovative, value-driven interactions that enhance consumer engagement (Eggenschwiler et al., 2024; Kaur et al., 2024). Overcoming physical constraints and fostering co-creation expands global reach and enables personalized brand interactions, key drivers of technology adoption (Marsh and Hocevar, 1985; Payal et al., 2024).

Past studies highlight that consumers are likelier to adopt innovations that offer distinct advantages over existing solutions (Rogers, 2003; Huang et al., 2023). Research in digital environments confirms that perceived relative advantage strengthens consumer confidence and adoption behavior (Opazo-Basáez et al., 2022; Parasuraman and Grewal, 2000). Thus,

H1d. Perceived relative advantage positively influences consumer attitudes toward the metaverse.

4.1.5. Combined influence of reasons for adoption

These "reasons for" adoption, as supported by qualitative findings and extant literature, collectively shape consumer attitudes. BRT shows positive drivers such as haptic imagery, convenience, trend affinity, and relative advantage synergistically enhance favorable attitudes toward new technologies (Claudy et al., 2015; Pillai and Sivathanu, 2020; Gupta and Arora, 2017). These factors position the metaverse as a transformative platform capable of addressing consumers' engagement, efficiency, and innovation needs.

H1. "Reasons for" adoption (haptic imagery, convenience, trend affinity, and relative advantage) positively influences consumer attitudes toward the metaverse.

4.2. Reasons against the adoption of the metaverse

While the metaverse holds significant potential, several concerns hinder its widespread adoption. Grounded in BRT, "reasons against" adoption represent negative factors that impede consumers' attitudes and behaviors toward adopting new technologies. Based on the qualitative findings and extant literature, four critical barriers were identified: privacy concerns, risk of malfunction, financial problems, and perceived performance risk. These obstacles reflect functional and psychological apprehensions that negatively influence consumers' attitudes toward the metaverse (Zhao et al., 2023a,b; Choi et al., 2018; Xu et al., 2013).

4.2.1. Privacy concerns

Privacy concerns emerged as a predominant barrier in the qualitative phase. Respondents frequently expressed apprehensions about how personal data is collected, stored, and used in metaverse platforms. Privacy concerns pose a significant barrier to metaverse adoption, driven by apprehensions about collecting, storing, and misusing personal data and unauthorized access to digital assets (Zhao et al., 2023a, b; Vimalkumar et al., 2021). In the metaverse, these fears are amplified due to the extensive data tracking and analytics required for immersive experiences, leading to negative perceptions and reduced consumer trust (Choi et al., 2018; Xu et al., 2013). Studies highlight that privacy concerns hinder engagement, with risks such as personal information leakage, scams, and stolen virtual assets particularly relevant in the metaverse context (Dwivedi et al., 2022). This aligns with e-commerce and virtual reality research evidence, where heightened privacy fears negatively affect consumer behavior and technology adoption (Pillai et al., 2024). Thus,

H2a. Privacy concerns negatively influence consumer attitudes toward the metaverse

4.2.2. Risk of malfunction

The risk of malfunction was another prominent barrier identified in the qualitative study. This fear of malfunction of Metaverse stems from its capacity to impact the user experience since user experience is a key determinant of technology adoption and engagement (Dwivedi et al., 2022). It has been observed that new technology makes users hesitant to participate. The reason for hesitation is fear of system failures, glitches, and software performance issues (Zhang et al., 2024). Users of Metaverse fear performance instability, such as lag or system crashes, which could disrupt their experience and immersion in the platform, posing a significant barrier to widespread adoption (Dwivedi et al., 2022). Past studies also note that when users perceive technology as unreliable, it

negatively impacts trust, engagement, and adoption (Featherman et al., 2021; Huang et al., 2023). Therefore, the authors propose the following hypothesis,

H2b. Risk of malfunction negatively influences consumer attitudes toward the metaverse.

4.2.3. Financial concerns

Financial concerns remain a critical barrier to metaverse adoption, with users perceiving significant costs associated with accessing and maintaining the required technologies, including VR headsets, software subscriptions, and digital assets (Featherman et al., 2021). Ongoing expenses for maintenance, system upgrades, and integration with existing technologies further exacerbate these concerns, especially when the long-term benefits and return on investment are uncertain (Mckinsey's et al., 2022). Such financial apprehensions are consistent with findings that perceived cost burdens negatively impact technology adoption, as consumers are often deterred by high upfront investments and recurring expenses (Vimalkumar et al., 2021). Additionally, perceiving the metaverse as a luxury rather than an essential service compounds these hesitations, particularly for users wary of hidden fees or unpredictable costs associated with its adoption (Kumar et al., 2024). Therefore,

H2c. Financial concerns negatively influence consumer attitudes toward the metaverse

4.2.4. Perceived performance risk

Perceived performance risk is a key barrier to metaverse adoption, as users doubt the platform's ability to deliver consistent, reliable, and high-quality experiences, which is critical for immersive technologies (Stone & Grønhaug, 1993; Yaqoob et al., 2023). Technical challenges like latency, system crashes, and integration issues amplify concerns, deterring engagement and reducing trust (Lowe, 2010; Sweeney et al., 1999). Past studies highlight that perceived performance risk significantly reduces technology adoption, as users associate system failures with wasted time, effort, and financial investment (Mitchell, 1999). Research in virtual and digital environments confirms that early-stage technologies often face skepticism due to perceived instability, negatively impacting adoption intentions (Yaqoob et al., 2023; Sweeney et al., 1999). Additionally, studies on technology acceptance models indicate that users are less likely to adopt innovations when they anticipate performance issues that disrupt usability and satisfaction (Van et al., 2020; Huang et al., 2023). Therefore,

H2d. Perceived performance risk negatively influences consumer attitudes toward the metaverse.

4.2.5. Combined influence of reasons against adoption

These "reasons against" adoption collectively undermine consumer attitudes, as evidenced in the qualitative findings and BRT's support. Negative drivers such as privacy concerns, risk of malfunction, financial matters, and perceived performance risks serve as significant barriers to adopting new technologies (Zhang et al., 2025; Hwang and Choe, 2019). These factors diminish consumer trust and confidence in the metaverse, ultimately impeding its adoption potential.

H2. "Reasons against" adoption (privacy concerns, risk of malfunction, financial problems, and perceived performance risk) negatively influence consumer attitudes toward the metaverse.

4.3. Values leading to the attitude formation toward metaverse

Values significantly impact how consumers feel about new technologies like the metaverse. In the context of technology readiness theory, these values are categorized as motivators, such as optimism and innovativeness, and inhibitors, such as discomfort and insecurity. These factors directly impact consumers' 'reason for' and 'reason against' using the metaverse technology, influencing their attitudes and results in the metaverse adoption intention.

4.3.1. Motivators of technology readiness

Motivators are constructive forces that encourage people to embrace the Metaverse, such as optimism and innovation. According to Parasuraman and Grewal (2000) and Jan et al. (2023), innovation is the willingness to adapt, whereas optimism is the hope for favorable results. Based on earlier research, these motivators enhance consumers' reactions to the Metaverse's haptic imagery (HI), convenience (CON), trend affinity (TA), and relative advantage (RA) (Sahu et al., 2020; Parasuraman and Colby, 2015). For instance, optimistic individuals will likely have a more favorable perception of the Metaverse as a seamless platform that provides more excellent value and satisfaction. On the other hand, innovative consumers are likely to be interested in the new and unique aspects of the Metaverse.

Hence, the following overarching hypothesis and sub-hypotheses are proposed:

H3. Motivators comprise innovativeness and optimism from technology readiness, which positively influences the formation of a 'reason for' adopting metaverse technology.

- **H3a**: Optimism positively influences haptic imagery (HI).
- **H3b**: Optimism positively influences convenience (CON).
- **H3c**: Optimism positively influences trend affinity (TA).
- **H3d**: Optimism positively influences relative advantage (RA).
- **H3e**: Innovativeness positively influences haptic imagery (HI).
- **H3f**: Innovativeness positively influences convenience (CON).
- **H3g**: Innovativeness positively influences trend affinity (TA).
- **H3h**: Innovativeness positively influences relative advantage (RA).

4.3.2. Inhibitors of technology readiness

On the other hand, there are inhibitors such as insecurity and discomfort that threaten the adoption of the Metaverse. Insecurity is linked toissues such as privacy, reliability, and data security, while discomfort is the perception of discomfort with the technology, which results in feelings of frustration or even avoidance (Blut and Wang, 2020; Jan et al., 2023). These inhibitors increase the "arguments against" adoption, for instance, privacy concerns (PC), risk of malfunction (RM), financial concerns (FC), and perceived performance risk (PPR). For example, those whoare insecure will highlight data privacy and system reliability as significant challenges. At the same time, the discomfort affecters may have problems with the technical complexity of the metaverse platforms' use. Thus, the following overarching hypothesis and sub-hypotheses are proposed:

H4. Inhibitors of technology readiness (insecurity and discomfort) positively influence the formation of "reasons against" metaverse adoption.

- **H4a**: Insecurity positively influences privacy concerns (PC).
- **H4b**: Insecurity positively influences the risk of malfunction (RM).
- **H4c**: Insecurity positively influences financial concerns (FC).
- **H4d**: Insecurity positively influences perceived performance risk (PPR).
- **H4e**: Discomfort positively influences privacy concerns (PC).
- **H4f**: Discomfort positively influences the risk of malfunction (RM).
- **H4g**: Discomfort positively influences financial concerns (FC).
- **H4h**: Discomfort positively influences perceived performance risk (PPR).

4.3.3. Values and attitude formation

The motivators and inhibitors of technology readiness alsodirectly impact consumer attitudes toward the metaverse. Optimism and

innovativeness foster positive attitudes by enhancing the perceived quality, functional value, and satisfaction derived from the metaverse. Conversely, insecurity and discomfort hinder attitude formation by amplifying apprehensions and reducing trust in the platform (Parasuraman and Colby, 2015; Blut and Wang, 2020).

Hence, the following hypotheses are proposed:

H5. Motivators of technology readiness positively influence consumer attitudes toward the metaverse.

- **H5a**: Optimism positively influences consumer attitudes toward the metaverse (ATT).
- **H5b**: Innovativeness positively influences consumer attitudes toward the metaverse (ATT).

H6. Inhibitors of technology readiness negatively influence consumer attitudes toward the metaverse.

- **H6a**: Insecurity negatively influences consumer attitudes toward the metaverse (ATT).
- **H6b**: Discomfort negatively influences consumer attitudes toward the metaverse (ATT).

4.4. Attitude towards the metaverse and adoption intention

Attitude is a key determinant of consumer behavior, which is the consumers' valence or expectation towards a particular behavior (Kasilingam and Krishna, 2022; Smith and Paladino, 2010). One of the most basic ideas in behavioral research is that attitudes influence how people make decisions, set goals, and even adopt new behaviors. The BRT states that the distinctions between the "reasons for" and "reasons against" arguments for adoption determine attitudes. Other theories that have also been used include the Theory of Planned Behavior (TPB; Ajzen, 1991), the Technology Acceptance Model (TAM; Davis, 1989), and the Theory of Reasoned Action (TRA; Fishbein and Ajzen, 1975) have also proven that attitudes have a direct relationship with adoption intentions. For instance, research in the context of smartphone chatbots (Kasilingam, 2020), organic foods (Tandon et al., 2020), e-waste recycling programs (Dhir et al., 2021), and micro wind turbines (Claudy et al., 2015) has established positive associations between attitudes formation and its leading impact on adoption intentions and engaging with such immersive technologies. It is evident from the above studies that attitude mediates the relationship between consumers' motivations and their actual behavior-related outcomes. Despite the strength of both these variables being well documented, they are scarcely explored within the metaverse literature, stating the significant gaps (Yu, 2024; Natarajan et al., 2024). Metaverse is considered novel in the entire digital ecosystem. It is inherited with unique challenges and uncertainties that need a comprehensive understanding of how attitudes are formed and how they impact the metaverse adoption intentions. Since the metaverse is immersive and dynamic, it is important to determine consumers' attitudes and how they translate into intentions to interact with this new technology. Based on the qualitative findings and existing theoretical frameworks, this study hypothesizes the following relationship:

H7. Attitude towards the metaverse will positively impact metaverse adoption intention.

4.5. Attitude towards the metaverse and engagement with the metaverse

Engagement has been defined in the literature as "the quality of experience typified by the consumer's cognitive, affective, and behavioral investment while interacting in the virtual environment" (Flavián et al., 2019; Hollebeek et al., 2014). Customer engagement (CE), discussed in academic literature since 2006, gained prominence after 2010, with varied definitions but no consensus. Scholars like Van Doorn et al. (2010) approached CE from behavioral and psychological perspectives, respectively, while Vivek et al. (2012) emphasized customer participation in a brand's activities. Collectively, CE aims to influence desired customer behaviors like positive brand attitudes, purchase intentions, and loyalty.

In the metaverse, customer engagement reflects how users interact with brands, technology, and others in an immersive digital space. It is a key driver of meaningful brand-consumer relationships, especially within interactive metaverse frameworks (Mollen and Wilson, 2010; Dwivedi et al., 2022).

Both academics and practitioners view engagement on immersive platforms as crucial (Flavián et al., 2019; Rasool et al., 2020; Arya et al., 2024). It fosters trust, satisfaction, loyalty, and perceived service quality (Kulikovskaja et al., 2023). For service brands, engagement is vital, enabling creative, interactive, and personalized consumer experiences (Kumar et al., 2019).

Despite its importance, limited research has examined engagement as a dependent variable, especially in the context of the metaverse (Ghali et al., 2024). Previous studies have highlighted the need to explore how consumer attitudes influence engagement in this unique virtual ecosystem (Payal et al., 2024). A positive attitude toward the metaverse will likely encourage active behavioral involvement, driving more significant interaction with brands and other users. By enhancing the consumer's emotional and cognitive connection to the platform, favorable attitudes translate into higher customer engagement.

Thus, the following hypothesis is proposed:

H8. Attitude towards the metaverse will positively impact customer engagement in the metaverse.

5. Quantitative study

5.1. Measurement of constructs

To test the hypotheses, we deployed an offline survey that was comprised of three parts. The first section briefly introduced the research objective to the participants, followed by the measures sourced from prior literature. In line with our review, we define the metaverse as a shared virtual environment that merges virtual, augmented, and mixed reality – to create immersively (e.g., work or leisure) experiences to market their products (Dwivedi et al., 2023; Singla et al., 2023).

The questionnaire was prepared following a literature review. We deployed established measures to gauge Adoption Intention (AI) and Attitude Towards the Metaverse (ATT), and they were measured by utilizing the items adapted from Fishbein and Ajzen (1975), Gupta and Arora (2017), and Bezbaruah and Trivedi (2020). Relative advantage has been measured by utilizing three items taken from Claudy et al. (2015). Three and four questions from Huang and Liao (2017) were used to measure trend affinity and haptic imagery. Four items were used from Malodia et al. (2022) to evaluate convenience. Financial concerns were measured using three items borrowed from Delgosha and Hajiheydari (2020).

The risk of malfunction was measured using three items from Lowe (2010) and Brack and Benkenstein (2014). Four items from Inman and Nikolova (2017) were used to measure privacy concerns. Eight items from Hollebeek et al. (2014) were used to measure user engagement on the metaverse platform. A pilot was conducted before the primary survey to ensure the respondents' understanding of the questionnaire wording. All items were rated on seven-point Likert scales ranging from "1" (strongly disagree) to "7" ("strongly agree"), as shown in Table 4.

A purposive sampling technique was used for data collection. The target population was customers with relevant experience using virtual environments such as AltspaceVR, Decentraland, and other blockchain-based platforms. These participants were identified and recruited from

Table 4
Scales.

Variable	Number of Items	Source
Adoption Intention (AI)	3	Fishbein and Ajzen (1975), Gupta and Arora (2017)
Attitude Towards the Metaverse (ATT)	4	Bezbaruah and Trivedi (2020)
Customer Engagement (ENG)	4	Hollebeek et al. (2014)
Convenience (CON)	4	Malodia et al. (2022)
Discomfort (DIS)	8	Parasuraman and Colby (2015); Jan et al. (2023)
Financial Concerns (FC)	4	Delgosha and Hajiheydari (2020)
Haptic Imagery (HI)	4	Huang and Liao (2017)
Innovativeness (INN)	4	Parasuraman and Colby (2015); Jan et al. (2023)
Insecurity (INS)	3	Parasuraman and Colby (2015); Jan et al. (2023)
Optimism (OPT)	6	Parasuraman and Colby (2015); Jan et al. (2023)
Perceived Performance Risk (PPR)	5	Lowe (2010); Brack and Benkenstein (2014)
Privacy Concerns (PC)	3	Inman and Nikolova (2017)
Relative Advantage (RA)	3	Claudy et al. (2015); Gupta and Arora (2017)
Risk of Malfunction (RM)	4	Huang et al. (2023)
Trend Affinity (TA)	3	Huang et al. (2023)

platforms like Discord and Reddit through communities like r/metaverse and r/metaverse_blockchain, which are dedicated to the metaverse. These platforms were considered suitable for accessing respondents familiar with the metaverse as they are frequently used by tech-savvy individuals and early adopters who engage with immersive digital environments. The communities on these platforms typically host active and niche-specific discussions among individuals with a demonstrated interest in the metaverse, thereby ensuring a representative population. As shown in Table 5, the survey sample reflects the demographic profile of metaverse users, as stated by Delloite report. Respondents were asked the following screening questions: 1) “Have you ever interacted with or used any metaverse-related technologies or platforms?” 2) “Do you intend to use metaverse in the coming future (next 3 months)?" Only individuals who answered “yes” to this question were requested to continue with the primary survey.

The sample consisted of 882 respondents. Following data collection, responses were screened to eliminate incomplete and inconsistent submissions. After this review, a final sample of 771 respondents was retained for analysis, ensuring the robustness and reliability of the findings. The demographic details of the final respondents’ are provided in Table 5.

Table 5
Sample demographics for the quantitative study.

Gender	Number of respondents	Percentage
Male	417	52 %
Female	339	44 %
Prefer Not to Say	15	4 %
Age Group		
Gen Z (born 1997–2012)	604	78.4 %
Millennials (born 1981–1996)	167	21.6 %
Income (USD)		
>10,000	305	39.55 %
10,001–15,000	206	26.71 %
15,001–20,000	169	21.91 %
>20,000	91	11.8 %
Highest Qualification		
Undergraduate	409	53 %
Postgraduate	316	41 %
Above Postgraduation	46	6 %
Usage of AltspaceVR, Decentraland/Any other metaverse		
Once (Non-Regular)	84	16 %
Two to Five times (Non Regular)	323	42 %
Once a week (Regular)	94	12 %
More than once a week (Regular)	231	30 %

This table outlines the distribution of participants by gender, age, and technology usage experience, highlighting the diversity within the sample that may influence attitudes toward metaverse adoption.

5.2. Results

Convergent validity was assessed to ensure that the constructs measured within the study reflect the same underlying theoretical concept. As presented in Table 6, the indicators for each construct achieved high loadings, all exceeding the threshold of 0.70, which indicates adequate convergent validity (Hair et al., 2017). The AVE values for each construct also met the criterion of more significant than 0.50, further confirming convergent validity. Specifically, constructs such as Haptic Imagery (AVE = 0.761), Optimism (AVE = 0.782), and Insecurity (AVE = 0.758) demonstrated strong convergent validity, indicating that these constructs effectively represent their respective domains.

Discriminant validity was evaluated using “the Fornell-Larcker criterion and the heterotrait-monotrait (HTMT) ratio. As per Table 7, the square roots of the AVE for each construct are presented along the diagonal, showing that they are greater than the correlations between constructs. For example, the square root of AVE for” Privacy Concerns (P.C.) is 0.867, which exceeds its correlation with Haptic Imagery (HI) at 0.498, thus confirming discriminant validity. Additionally, the HTMT ratio provided a supplementary discriminant validity assessment, as shown in Table 7. Every HTMT value fell below the 0.85 threshold that is advised (Henseler et al., 2015), further substantiating that constructs are distinct. For instance, the HTMT value between Innovativeness (INN) and Engagement (ENG) was 0.708, indicating sufficient separation between these constructs.

To address concerns related to Common Method Bias (CMB) and Common Method Variance (CMV), this study implemented both procedural and statistical remedies to enhance the validity of the findings. Procedurally, respondents were assured of anonymity and confidentiality, thereby minimizing social desirability bias. The survey design also incorporated separation between predictor and criterion variables to reduce common rater effects that help in mitigating common scale method bias. To minimize systematic response bias, the sequence of the survey items were randomized for each respondent, hence minimizing the likelihood of order effects influencing the responses. Statistical remedies included Harman’s Single-Factor Test, which revealed that no single factor accounted for most of the variance, with the most prominent factor explaining only 37.23 %. This is well below the 50 % threshold, indicating that CMB was not a significant issue (Podsakoff et al., 2003). The results of the Common Latent Factor (CLF) approach indicated that the variance explained by the common factor remained below 25 %, further suggesting minimal method bias (Chin et al., 2013). Variance Inflation Factor (VIF) analysis also showed that all predictor constructs had VIF values below 3.3, ensuring that multicollinearity and CMV were not significant concerns (Kock, 2015). Following these procedural remedies and statistical assessments, the influence of CMB and CMV in this study is negligible, thereby ensuring the robustness of the findings. To assess non-response bias, we conducted a wave analysis comparing early (first 50 %) and late (last 50 %) respondents across demographics (age, gender, income, education) and key study variables (attitudes, adoption intention, engagement). Independent sample t-tests and chi-square tests showed no significant differences ($p > 0.05$), indicating minimal non-response bias (Armstrong and Overton, 1977; Lindner et al., 2001).

Several fit indices were examined to evaluate the structural model’s goodness of fit. The Standardized Root Mean Square Residual (SRMR) was 0.047, below the recommended threshold of 0.08, indicating a good model fit (Hu and Bentler, 1999). The Normed Fit Index (NFI) yielded a value of 0.797, which is lower than the conventional benchmark of 0.90

Table 6
Convergent validity assessment.

Variable	Item ID	Loadings	Average Loading	Cronbach's Alpha	Composite Reliability (CR)	Average variance extracted (AVE)	Pilot Study Cronbach's Alpha
Adoption Intention	AI1	0.924	0.933	0.926	0.953	0.871	0.871
	AI2	0.944					
	AI3	0.931					
Attitude Towards the Metaverse	ATT1	0.813	0.847	0.869	0.911	0.718	0.718
	ATT2	0.865					
	ATT3	0.871					
	ATT4	0.839					
Convenience	CON1	0.883	0.858	0.88	0.918	0.736	0.736
	CON2	0.835					
	CON3	0.877					
	CON4	0.835					
Discomfort	DIS1	0.675	0.748	0.74	0.837	0.563	0.563
	DIS2	0.807					
	DIS3	0.739					
	DIS4	0.770					
Brand Engagement	ENG1	0.627	0.744	0.886	0.911	0.565	0.565
	ENG2	0.694					
	ENG3	0.815					
	ENG4	0.862					
	ENG5	0.860					
	ENG6	0.844					
	ENG7	0.671					
	ENG8	0.582					
Financial Concerns	FC1	0.832	0.739	0.724	0.830	0.558	0.558
	FC2	0.845					
	FC3	0.649					
	FC4	0.631					
Haptic Imagery	HI1	0.673	0.789	0.808	0.873	0.635	0.635
	HI2	0.730					
	HI3	0.890					
	HI4	0.861					
Innovativeness	INN1	0.888	0.835	0.857	0.904	0.703	0.703
	INN2	0.764					
	INN3	0.781					
	INN4	0.908					
Insecurity	INS1	0.835	0.822	0.762	0.864	0.68	0.68
	INS2	0.887					
	INS3	0.745					
Optimism	OP1	0.844	0.752	0.847	0.889	0.574	0.574
	OP2	0.792					
	OP3	0.809					
	OP4	0.749					
	OP5	0.763					
	OP6	0.556					
Privacy Concerns	PC1	0.842	0.899	0.941	0.955	0.809	0.809
	PC2	0.893					
	PC3	0.942					
	PC4	0.927					
	PC5	0.889					
Perceived Performance Risk	PPR1	0.903	0.882	0.859	0.914	0.78	0.78
	PPR2	0.895					
	PPR3	0.847					
Relative Advantage	RA1	0.879	0.886	0.863	0.916	0.784	0.784
	RA2	0.878					
	RA3	0.900					
Risk of Malfunction	RM1	0.894	0.871	0.895	0.927	0.76	0.76
	RM2	0.864					
	RM3	0.854					
	RM4	0.871					
Trend Affinity	TA1	0.835	0.841	0.794	0.879	0.709	0.709
	TA2	0.821					
	TA3	0.866					

(Bentler and Bonett, 1980). However, recent literature suggests that NFI is highly sensitive to sample size and model complexity and often underestimates model fit in complex structural models with multiple latent constructs (Hair et al., 2019; Henseler et al., 2016). In such cases, SRMR and other fit indices (such as the Comparative Fit Index - CFI and Tucker-Lewis Index - TLI) provide a more comprehensive assessment of model fit (Henseler et al., 2016). Prior research has also indicated that NFI tends to decrease in models with high degrees of freedom, especially when using Partial Least Squares Structural Equation Modeling (PLS-SEM) (Henseler et al., 2015). Given that SRMR falls within the acceptable range and other model fit indices suggest an adequate fit, the structural model is deemed acceptable based on established recommendations in PLS-SEM research (Hair et al., 2017).

After assessing the convergent and discriminant validity, common method bias, and model fit, the structural model was evaluated. The results supported Hypothesis 1 (H1), indicating that reasons for

Table 7
Discriminant validity assessment.

	AI	ATT	CON	DIS	ENG	FC	HI	INN	INS	OPT	PC	PPR	RA	RM	TA
AI	**0.933**	0.810	0.822	0.613	0.124	0.351	0.482	0.551	0.072	0.534	0.255	0.176	0.576	0.155	0.673
ATT	0.729	**0.847**	0.888	0.739	0.274	0.423	0.598	0.675	0.142	0.651	0.305	0.232	0.637	0.200	0.549
CON	0.741	0.786	**0.752**	0.895	0.241	0.448	0.671	0.639	0.129	0.538	0.383	0.254	0.806	0.157	0.759
DIS	0.554	0.649	0.794	**0.858**	0.239	0.339	0.748	0.573	0.067	0.507	0.251	0.218	0.894	0.182	0.727
ENG	0.085	0.189	0.157	0.181	**0.749**	0.612	0.273	0.268	0.629	0.234	0.420	0.529	0.161	0.599	0.273
FC	0.282	0.331	0.357	0.275	0.448	**0.746**	0.227	0.406	0.551	0.321	0.747	0.682	0.403	0.539	0.474
HI	0.440	0.540	0.580	0.665	0.198	0.158	**0.794**	0.429	0.115	0.455	0.254	0.168	0.666	0.179	0.467
INN	0.500	0.591	0.571	0.510	0.186	0.330	0.387	**0.838**	0.265	0.634	0.470	0.327	0.572	0.156	0.491
INS	0.023	0.104	0.048	0.000	0.485	0.403	0.064	0.171	**0.824**	0.237	0.588	0.591	0.114	0.631	0.170
OPT	0.473	0.563	0.462	0.442	0.141	0.237	0.402	0.535	0.143	**0.758**	0.159	0.344	0.583	0.181	0.557
PC	0.233	0.271	0.328	0.214	0.326	0.597	0.218	0.405	0.483	0.138	**0.882**	0.463	0.389	0.415	0.264
PPR	0.166	0.204	0.168	0.188	0.458	0.564	0.172	0.285	0.511	0.301	0.424	**0.899**	0.248	0.682	0.285
RA	0.514	0.550	0.704	0.780	0.127	0.324	0.570	0.497	0.047	0.494	0.324	0.225	**0.886**	0.194	0.799
RM	0.143	0.175	0.113	0.154	0.509	0.431	0.125	0.123	0.532	0.115	0.364	0.634	0.172	**0.871**	0.248
TA	0.574	0.462	0.631	0.603	0.178	0.367	0.400	0.423	0.115	0.459	0.210	0.249	0.657	0.212	**0.841**

Note: Elements with the bold font on the diagonal indicate $\sqrt{}$the AVE of the latent variable. The HTMT ratio appears above $\sqrt{}$AVE, and Pearson's correlation coefficients are listed below. Fit indices: standardized root mean square residual (SRMR); = 0.047, normed fit index (NFI) = 0.797.

adoption significantly influence attitudes toward the metaverse (ATT). Specifically, Haptic Imagery (HI) showed a positive and significant relationship with ATT, with a beta coefficient of 0.152, a t-value of 2.915, and a p-value of 0.004, suggesting that immersive experiences are crucial in fostering positive attitudes toward the metaverse. Furthermore, Convenience (CON) exhibited an even more substantial effect on ATT, with a beta of 0.347, a t-value of 5.062, and a p-value of <0.001, highlighting that participants perceived convenience as a significant factor in shaping their attitudes. However, Trend Affinity (TA) presented a non-significant relationship ($\beta = 0.005$, $t = 0.080$, $p = 0.936$), indicating that the trendiness of technology does not directly contribute to positive attitudes. Additionally, Relative Advantage (RA) displayed a slight negative effect ($\beta = -0.081$, $t = 1.053$, $p = 0.293$), suggesting that participants did not perceive substantial advantages of the metaverse over traditional media.

Hypothesis 2 (H2) explored reasons against adoption, which showed no significant influence on ATT ($\beta = 0.027$, $t = 0.422$, $p = 0.673$). However, when examining the sub-factors, Privacy Concerns (PC) revealed a significant negative relationship with ATT ($\beta = -0.165$, $t = 2.515$, $p = 0.012$), which indicates that higher privacy concerns adversely affect attitudes toward the metaverse. Interestingly, Risk of Malfunction (RM) displayed a positive effect ($\beta = 0.125$, $t = 2.380$, $p = 0.017$), suggesting that while participants recognized risks, they were also aware of potential technological improvements that slightly mitigated their concerns. Financial Concerns (FC) and Perceived Performance Risk (PPR) exhibited minimal influence (FC: $\beta = 0.122$, $t = 1.488$, $p = 0.137$; PPR: $\beta = 0.013$, $t = 0.202$, $p = 0.840$), indicating that these factors were not significantly impactful in shaping attitudes.

The analysis confirmed Hypothesis 3 (H3), revealing that motivators significantly affect reasons for adoption. The results indicated strong positive relationships, with Optimism (OPT) demonstrating a robust positive influence on Haptic Imagery (HI) ($\beta = 0.287$, $t = 4.454$, $p < 0.001$) and Innovativeness (INN) also contributing positively ($\beta = 0.236$, $t = 3.735$, $p < 0.001$). These outcomes suggest that a positive outlook and a propensity for innovation enhance perceptions of immersive experiences in the metaverse.

Moreover, the analysis supported Hypothesis 4 (H4), indicating that inhibitors significantly impact reasons against adoption. Insecurity (INS) positively affected Privacy Concerns (PC) with a coefficient of $\beta = 0.377$, a t-value of 5.098, and $p < 0.001$, which suggests that feelings of insecurity strongly correlate with privacy apprehensions. Discomfort (DIS) also presented a substantial effect ($\beta = 0.329$, $t = 5.871$, $p < 0.001$), indicating that physical discomfort associated with technology exacerbates privacy concerns, thereby deterring potential adopters.

Hypothesis 5 (H5) proposed that motivators positively influence attitudes toward the metaverse (ATT). The PLS-SEM analysis supported this hypothesis, revealing a significant positive effect of motivators on ATT, with a path coefficient of $\beta = 0.402$, t-value of 7.614, and p-value <0.001. Among the motivators, both Optimism (OPT) and Innovativeness (INN) significantly contributed to a positive attitude toward the metaverse. Specifically, Optimism directly impacted ATT ($\beta = 0.268$, $t = 4.675$, $p < 0.001$), indicating that individuals with a positive outlook on technology were more inclined to view the metaverse favorably. Similarly, Innovativeness also significantly affected ATT ($\beta = 0.245$, $t = 4.902$, $p < 0.001$), suggesting that participants who identified as early adopters or technology enthusiasts were more likely to adopt favorable attitudes toward the metaverse.

In contrast, Hypothesis 6 (H6) posited that inhibitors would negatively influence attitudes toward the metaverse. However, the data did not support this hypothesis, as the relationship between inhibitors and ATT was minimal and statistically non-significant ($\beta = 0.001$, $t = 0.012$, $p = 0.990$). Neither Insecurity (INS) nor Discomfort (DIS) significantly impacted attitudes, with INS showing a near-zero influence ($\beta = 0.001$, $t = 0.017$, $p = 0.986$) and DIS also demonstrating a non-significant effect ($\beta = -0.028$, $t = 0.517$, $p = 0.605$). These findings indicate that contrary to expectations, concerns about insecurity and physical discomfort did not deter participants from forming positive attitudes toward the metaverse. This unexpected result suggests that, while inhibitors may contribute to reasons against adoption, they do not directly affect users' overall attitudes like motivators do.

Finally, the analysis highlighted that attitudes toward the metaverse significantly influence actual interaction (AI) and engagement (ENG), as confirmed in Hypothesis 7 (H7) and Hypothesis 8 (H8). The outcomes depicted a robust positive association between ATT & AI ($\beta = 0.662$, $t = 12.885$, $p < 0.001$), indicating that positive attitudes are critical for fostering user interaction with metaverse technologies. Similarly, ATT significantly impacted ENG ($\beta = 0.645$, $t = 12.505$, $p < 0.001$), emphasizing that favorable attitudes enhance user engagement within metaverse environments.

In order to understand how the demographic factors influence Attitude Toward the Metaverse (ATT) and its predictors, this study examined the impact of Metaverse Usage, Gender, Income Level, Education Level, and Generation on the strengths of the relationships hypothesized in the model through a multi-group bootstrapped analysis. The analysis presents significant insights into how different user groups perceive and engage with the metaverse.

Comparing the regular and non-regular metaverse users, optimism significantly influenced ATT for regular users ($\beta = 0.312$, $t = 4.496$, $p < 0.001$). However, this relationship was insignificant for non-regular users ($\beta = 0.197$, $t = 1.93$, $p > 0.05$). This suggests that optimistic individuals who frequently engage with the metaverse develop stronger positive attitudes, while those with limited exposure do not exhibit the

same enthusiasm. Similarly, innovativeness had a significant impact on ATT for regular users ($\beta = 0.248$, $t = 3.966$, $p < 0.001$) but was insignificant for non-regular users ($\beta = 0.171$, $t = 1.867$, $p > 0.05$), indicating that technology familiarity plays a key role in shaping metaverse adoption. Interestingly, privacy concerns significantly reduced ATT for non-regular users ($\beta = -0.282$, $t = 3.063$, $p < 0.01$), but this effect was insignificant for regular users ($\beta = 0.01$, $t = 0.096$, $p > 0.05$), suggesting that privacy apprehensions are a significant deterrent for those unfamiliar with the metaverse.

The analysis of gender differences in metaverse adoption revealed key variations in the strength of specific relationships. Financial concerns significantly influenced attitudes for males ($\beta = 0.368$, $t = 3.552$, $p < 0.001$) but were not significant for females, with a statistically significant difference in effect size. This suggests that financial constraints are more crucial in shaping male adoption decisions. Additionally, performance risk (PPR ATT) was significant only for males ($\beta = -0.192$, $t = 2.812$, $p < 0.05$), indicating that men perceive performance risks as a stronger deterrent. Furthermore, the influence of trend affinity (TA) on attitude (ATT) was significant for males ($\beta = 0.272$, $t = 1.991$, $p < 0.05$) but not for females, suggesting that males are more likely to develop favorable attitudes toward the metaverse based on trend-driven influences. Additionally, the effect of attitude on adoption intention (AI) was stronger for males ($\beta = 0.820$, $t = 20.875$, $p < 0.001$) compared to females ($\beta = 0.683$, $t = 15.461$, $p < 0.001$), indicating that attitude plays a more critical role in shaping metaverse adoption among male users.

Income level played a critical role in shaping attitudes toward the metaverse. Among higher-income individuals, optimism had a significant impact on ATT ($\beta = 0.361$, $t = 5.547$, $p < 0.001$), whereas the effect was insignificant for lower-income users ($\beta = 0.201$, $t = 1.772$, $p > 0.05$). This suggests that financial stability fosters a more positive outlook on digital adoption. Furthermore, financial concerns significantly reduced ATT for lower-income users ($\beta = 0.3$, $t = 2.647$, $p < 0.01$), but the effect was insignificant for higher-income individuals ($\beta = 0.214$, $t = 1.229$, $p > 0.05$), reinforcing the idea that financial constraints act as a stronger barrier for those with limited resources.

Education level also influenced metaverse adoption patterns. Among highly educated individuals, innovativeness significantly influenced ATT ($\beta = 0.289$, $t = 4.681$, $p < 0.001$), whereas the effect was insignificant for lower-educated individuals ($\beta = 0.227$, $t = 2.804$, $p > 0.05$). This suggests that higher education fosters greater openness to technological advancements. Conversely, discomfort had a more negative impact on ATT among lower-educated individuals ($\beta = -0.15$, $t = 1.221$, $p > 0.05$), while the effect was insignificant for highly educated individuals ($\beta = 0.019$, $t = 0.234$, $p > 0.05$). This finding indicates that individuals with lower education levels may struggle more with adopting complex digital technologies.

Finally, generational differences revealed that Gen-Z respondents were more engaged in the metaverse than Millennials. ATT had a significantly more substantial relationship with engagement (ENG) among Gen-Z users ($\beta = 0.885$, $t = 30.077$, $p < 0.001$) compared to Millennials ($\beta = 0.747$, $t = 26.056$, $p < 0.001$), indicating that Gen-Z users are more likely to actively participate in the metaverse once they develop positive attitudes. Additionally, privacy concerns significantly reduced ATT among Millennials ($\beta = -0.142$, $t = 2.017$, $p < 0.05$), but the effect was insignificant for Gen-Z participants ($\beta = -0.128$, $t = 1.774$, $p > 0.05$). This suggests that Millennials are generally more cautious about privacy risks, whereas Gen-Z users are comparatively less concerned.

This demographic analysis highlights that different user groups perceive and engage with the metaverse differently based on exposure, income, education, gender, and generational identity. Regular users, highly educated individuals, higher-income respondents, and Gen-Z participants exhibited stronger adoption tendencies. In contrast, non-regular users, lower-income individuals, lower-educated participants, and Millennials expressed more significant concerns about privacy, financial constraints, and technological discomfort. Furthermore, optimism and innovativeness played a critical role in shaping ATT across most groups, though the strength of these relationships varied significantly by demographics. For details, refer to Table 8.

6. Discussion

This study investigates the factors influencing consumer adoption of the metaverse by integrating BRT and TRI to develop a comprehensive framework that examines both the drivers and barriers to metaverse engagement. To enhance the validity and robustness of our findings, this study employed a sequential mixed-methods approach containing both qualitative insights from industry experts and quantitative survey data. The qualitative phase, with 48 expert interviews, identified key motivators, namely, trend affinity, gamification, and immersive engagement, as well as barriers, including privacy concerns, financial constraints, and performance risks. These qualitative themes were empirically tested in the quantitative phase using survey data from 771 Gen Z and Millennial consumers. Our findings indicate that optimism and innovativeness significantly influence adoption attitudes (H5, $p < 0.001$), especially among Gen Z consumers. Trend affinity was also a significant predictor of adoption attitudes (H1c, $p < 0.05$), but its influence was stronger for males than females. Financial concern's impact on attitude was significant for Millennials (H2c, $p < 0.05$) but not for Gen Z consumers, indicating a generational divide in cost sensitivity. While some industry experts from the qualitative field highlighted privacy concerns as a significant adoption barrier, the quantitative results showed that privacy concerns were not significant deterrents for a few user groups (H2a, $p > 0.05$), including young users. This raises the notion that younger consumers may be more willing to trade privacy for immersive digital experiences. However, performance risks were a significant barrier, especially for male users (H2d, $p < 0.05$), indicating the need for robust technical infrastructure to enhance user confidence. This integration of qualitative and quantitative insights strengthens our findings, demonstrating alignment between industry expectations and empirical evidence. Below, we further discuss these findings about the study's hypotheses and prior research.

6.1. Reasons for metaverse adoption

The results support BRT's assertion that individuals justify their behaviors based on perceived benefits (Westaby, 2005). Optimism (H3), Innovativeness (H4), and Perceived Advantages (H5) had statistically significant positive effects on metaverse adoption attitudes, reinforcing that technology enthusiasts and digital consumers are more inclined to engage with immersive digital platforms (Parasuraman, 2000; Yu, 2024). Optimism (H3) was a particularly strong driver, indicating that individuals who perceive technology as enhancing daily experiences are more likely to embrace the metaverse. This aligns with prior research suggesting that tech-savvy consumers, mainly Gen Z and Millennials, view immersive environments as extensions of their digital lifestyles (Deloitte, 2023; Payal et al., 2024). Similarly, innovativeness (H4) positively influenced adoption, particularly among early adopters seeking novel and immersive experiences (Adhini and Prasad, 2024). Additionally, perceived advantages (H5) were a major driver of adoption, reinforcing that elements such as haptic imagery, virtual commerce, and social connectivity significantly impact consumer attitudes (Park and Lim, 2023). Our qualitative insights further highlight that gamification and digital collectibility (e.g., NFTs, avatars, and digital goods) drive engagement, confirming previous studies on consumer immersion in virtual spaces (Mogaji et al., 2023).

6.2. Reasons against metaverse adoption

While negative reasoning (inhibitors) was expected to strongly impact attitudes, barriers such as insecurity and financial risks (H6) had a comparatively weaker influence. Privacy concerns, often cited as a

Table 8
SEM results.

H. No	Hypothesis	First Order Model	Second Order Model	Metaverse Usage			Gender			Income Level			Education Level			Generation		
				Regular	Non-regular	Difference	Male	Female	Difference	Lower Income	Higher Income	Difference	Upto Undergraduate	Postgraduate and Above	Difference	Millennials	Gen-Z	Difference
H1	Reasons For → ATT		0.42*** (7.256)															
H1a	HI → ATT	0.152** (2.915)		0.269** (2.977)	0.104 (1.305)	0.166 (1.375)	0.105 (1.288)	0.178ˆ (1.827)	0.074 (0.589)	0.212** (3.049)	0.178ˆ (1.66)	0.035 (0.285)	0.04 (0.182)	0.179** (2.756)	0.14 (0.801)	0.388 (1.547)	0.147* (2.398)	0.241 (1.383)
H1b	CON → ATT	0.347*** (5.062)		0.314* (2.404)	0.368*** (4.123)	0.054 (0.354)	0.428*** (4.033)	0.338* (2.445)	0.09 (0.527)	0.339*** (3.751)	0.345* (2.573)	0.006 (0.038)	0.544* (2.041)	0.357*** (4.745)	0.187 (0.912)	0.037 (0.113)	0.374*** (5.107)	0.336 (1.521)
H1c	TA → ATT	0.005 (0.08)		0.155 (1.198)	−0.087 (1.097)	0.242ˆ (1.686)	−0.124 (1.459)	0.148 (1.335)	0.272* (1.991)	0.088 (1.047)	−0.205* (2.105)	0.293* (2.281)	0.035 (0.134)	−0.021 (0.287)	0.056 (0.28)	0.158 (0.901)	−0.044 (0.577)	0.202 (1.174)
H1d	RA → ATT	−0.081 (1.053)		−0.175 (1.451)	−0.021 (0.203)	0.153 (0.953)	−0.100 (1.003)	−0.179 (1.313)	0.079 (0.480)	−0.218* (2.347)	0.111 (0.906)	0.329* (2.191)	0.1 (0.33)	−0.134ˆ (1.696)	0.234 (1.056)	−0.246 (0.982)	−0.075 (0.917)	0.171 (0.846)
H2	Reasons Against → ATT		0.027 (0.422)															
H2a	PC → ATT	−0.165* (2.515)		0.01 (0.096)	−0.282** (3.063)	0.292* (2.079)	−0.135 (1.568)	−0.187ˆ (1.655)	0.052 (0.376)	−0.245** (2.708)	−0.105 (0.991)	0.14 (1.013)	−0.418ˆ (1.861)	−0.142* (2.017)	0.276 (1.48)	−0.391* (1.961)	−0.128ˆ (1.774)	0.263 (1.53)
H2b	RM → ATT	0.125* (2.38)		0.028 (0.343)	0.148ˆ (1.774)	0.12 (1.008)	0.036 (0.466)	0.150ˆ (1.850)	0.113 (0.978)	0.125ˆ (1.791)	0.08 (0.725)	0.044 (0.355)	0.325 (1.46)	0.097ˆ (1.706)	0.229 (1.428)	−0.09 (0.556)	0.15* (2.369)	0.24 (1.636)
H2c	FC → ATT	0.122 (1.488)		0.043 (0.283)	0.238ˆ (1.708)	0.195 (0.943)	0.368*** (3.552)	−0.058 (0.423)	0.426** (2.533)	0.3** (2.647)	0.087 (0.653)	0.214 (1.229)	0.13 (0.532)	0.194ˆ (1.782)	0.063 (0.237)	0.69*** (3.44)	0.069 (0.687)	0.621** (2.848)
H2d	PPR → ATT	0.013 (0.202)		−0.079 (0.95)	−0.031 (0.409)	0.048 (0.422)	−0.192* (2.812)	0.100 (1.065)	0.292 ** (2.585)	−0.161* (2.14)	0.126 (1.6)	0.287** (2.597)	−0.057 (0.295)	−0.036 (0.588)	0.021 (0.132)	−0.235ˆ (1.708)	0 (0.002)	0.235ˆ (1.682)
H3	Motivators → Reasons For		0.576*** (12.123)															
H3a	OPT → HI	0.287*** (4.454)		0.336*** (3.313)	0.211* (2.376)	0.125 (0.927)	0.278*** (3.192)	0.315** (2.802)	0.037 (0.266)	0.231** (2.793)	0.341** (3.1)	0.11 (0.818)	0.433** (3.089)	0.259*** (3.564)	0.174 (0.996)	0.275 (1.606)	0.251*** (3.65)	0.023 (0.148)
H3b	OPT → CON	0.242*** (3.492)		0.341*** (3.335)	0.139 (1.538)	0.202 (1.483)	0.214ˆ (2.221)	0.297* (2.886)	0.083 (0.586)	0.169* (2.074)	0.313* (2.553)	0.144 (1.021)	0.498*** (4.466)	0.203** (2.633)	0.296 (1.641)	0.381** (2.932)	0.169* (2.154)	0.212 (1.297)
H3c	OPT → TA	0.303*** (4.064)		0.422*** (3.775)	0.25* (2.536)	0.172 (1.154)	0.309*** (3.414)	0.428*** (3.762)	0.118 (0.828)	0.225* (2.159)	0.465*** (4.897)	0.24 (1.639)	0.581*** (5.361)	0.281** (3.261)	0.3 (1.503)	0.306 (1.621)	0.322*** (4.17)	0.016 (0.089)
H3d	OPT → RA	0.295*** (4.442)		0.443*** (4.299)	0.185* (2.099)	0.259ˆ (1.915)	0.320*** (3.432)	0.372*** (3.911)	0.052 (0.386)	0.296*** (3.566)	0.377*** (3.307)	0.082 (0.596)	0.508*** (4.47)	0.292*** (3.807)	0.215 (1.194)	0.36* (2.545)	0.289*** (3.958)	0.071 (0.454)
H3e	INN → HI	0.236*** (3.735)		0.251* (2.291)	0.239** (2.754)	0.011 (0.081)	0.288** (3.065)	0.194ˆ (1.909)	0.094 (0.680)	0.227** (2.804)	0.254* (2.158)	0.027 (0.194)	0.062 (0.352)	0.268*** (3.754)	0.205 (1.152)	0.373* (2.381)	0.21** (2.774)	0.164 (0.991)
H3f	INN → CON	0.383*** (7.652)		0.371*** (4.185)	0.406*** (6.127)	0.034 (0.319)	0.380*** (5.340)	0.366*** (4.610)	0.014 (0.133)	0.407*** (6.95)	0.351*** (3.456)	0.056 (0.513)	0.107 (0.838)	0.427*** (7.585)	0.32* (2.311)	0.443*** (3.682)	0.37*** (6.3)	0.073 (0.573)
H3g	INN → TA	0.264*** (4.698)		0.223* (2.263)	0.291*** (3.917)	0.068 (0.564)	0.269*** (3.383)	0.192* (2.069)	0.078 (0.639)	0.285*** (3.762)	0.184ˆ (1.904)	0.1 (0.831)	0.22ˆ (1.88)	0.25*** (3.843)	0.03 (0.192)	0.341* (2.269)	0.24*** (3.74)	0.101 (0.698)
H3h	INN → RA	0.342*** (6.172)		0.214* (2.203)	0.437*** (6.07)	0.222ˆ (1.886)	0.304*** (3.729)	0.309*** (3.753)	0.05 (0.040)	0.366*** (5.715)	0.249* (2.284)	0.117 (0.981)	0.142 (0.922)	0.355*** (5.667)	0.213 (1.363)	0.463*** (4.084)	0.298*** (4.681)	0.165 (1.224)
H4	Inhibitors → Reasons Against		0.689*** (14.077)															
H4a	INS → PC	0.377*** (5.098)		0.268* (2.344)	0.488*** (5.735)	0.221 (1.591)	0.405*** (3.700)	0.366*** (3.876)	0.038 (0.257)	0.437*** (4.568)	0.328** (3.243)	0.109 (0.774)	0.021 (0.08)	0.421*** (5.645)	0.401* (1.984)	0.505*** (3.396)	0.352*** (4.603)	0.153 (0.931)
H4b	INS → RM	0.372*** (6.282)		0.364*** (3.601)	0.398*** (5.739)	0.034 (0.286)	0.403*** (4.884)	0.339*** (3.529)	0.064 (0.506)	0.401*** (4.801)	0.356*** (4.292)	0.046 (0.379)	0.143 (0.773)	0.411*** (6.78)	0.268ˆ (1.686)	0.294* (2.35)	0.388*** (6.054)	0.094 (0.684)
H4c	INS → FC	0.197* (2.377)		0.244ˆ (1.859)	0.254* (2.211)	0.01 (0.057)	0.192 (1.609)	0.347** (2.793)	0.154 (0.889)	0.249* (2.422)	0.228ˆ (1.692)	0.02 (0.123)	−0.342 (0.972)	0.308*** (3.605)	0.65** (2.644)	0.517*** (4.219)	0.157 (1.602)	0.36ˆ (1.831)
H4d	INS → PPR	0.425*** (6.359)		−0.004 (0.041)	0.032 (0.341)	0.036 (0.261)	0.008 (0.078)	0.031 (0.311)	0.023 (0.163)	0.044 (0.427)	−0.027 (0.305)	0.071 (0.499)	0.068 (0.313)	0.021 (0.287)	0.047 (0.252)	−0.001 (0.009)	0.009 (0.118)	0.01 (0.064)

(*continued on next page*)

Table 8 (*continued*)

H. No	Hypothesis	First Order Model	Second Order Model	Metaverse Usage			Gender			Income Level			Education Level			Generation		
				Regular	Non-regular	Difference	Male	Female	Difference	Lower Income	Higher Income	Difference	Upto Undergraduate	Postgraduate and Above	Difference	Millennials	Gen-Z	Difference
H4e	DIS → PC	0.277*** (4.452)		0.312** (2.991)	0.228** (3.105)	0.084 (0.682)	0.146 (1.426)	0.444*** (6.448)	0.299* (2.292)	0.207* (2.487)	0.349*** (3.96)	0.142 (1.154)	0.507* (2.242)	0.256*** (3.937)	0.25 (1.418)	0.344* (2.321)	0.216** (3.145)	0.129 (0.849)
H4f	DIS → RM	0.329*** (5.871)		0.28** (2.857)	0.368*** (5.562)	0.088 (0.774)	0.274** (3.167)	0.393*** (4.972)	0.119 (0.996)	0.275*** (3.511)	0.393*** (4.951)	0.118 (1.038)	0.49** (2.839)	0.309*** (5.202)	0.18 (1.175)	0.531*** (6.177)	0.266*** (3.879)	0.265ˆ (1.926)
H4g	DIS → FC	0.272*** (4.485)		0.381*** (4.284)	0.289** (3.098)	0.093 (0.699)	0.354*** (3.490)	0.379*** (5.018)	0.024 (0.185)	0.366*** (3.542)	0.347*** (4.509)	0.019 (0.138)	0.58** (2.846)	0.319*** (4.538)	0.261 (1.435)	0.319** (2.728)	0.296*** (3.625)	0.022 (0.133)
H4h	DIS → PPR	0.117 (1.592)		0.485*** (4.94)	0.363*** (3.807)	0.123 (0.881)	0.409*** (4.133)	0.448*** (4.472)	0.039 (0.274)	0.373*** (3.482)	0.487*** (5.28)	0.114 (0.772)	0.367 (1.529)	0.417*** (5.524)	0.05 (0.25)	0.627*** (5.342)	0.358*** (4.36)	0.268 (1.601)
H5	Motivators → ATT		0.402*** (7.614)															
H5a	OPT → ATT	0.268*** (4.675)		0.197ˆ (1.93)	0.312*** (4.496)	−0.115 (0.968)	0.310*** (4.545)	0.239* (2.435)	0.071 (0.616)	0.361*** (5.547)	0.161 (1.623)	0.201ˆ (1.772)	0.189 (0.811)	0.267*** (4.538)	0.078 (0.469)	0.452** (3.03)	0.268*** (4.282)	0.184 (1.299)
H5b	INN → ATT	0.245*** (4.902)		0.171ˆ (1.867)	0.248*** (3.966)	0.077 (0.722)	0.175** (2.660)	0.319*** (3.775)	0.144 (1.370)	0.217*** (3.317)	0.289** (2.958)	0.071 (0.63)	0.163 (1.018)	0.251*** (4.324)	0.088 (0.592)	0.167 (1.272)	0.283*** (4.633)	0.116 (0.86)
H6	Inhibitors → ATT		0.001 (0.012)															
H6a	INS → ATT	0.001 (0.017)		−0.009 (0.109)	0.02 (0.25)	0.029 (0.248)	0.019 (0.235)	−0.042 (0.419)	0.061 (0.484)	−0.088 (1.144)	0.113 (1.514)	0.201ˆ (1.827)	0.029 (0.114)	−0.024 (0.401)	0.053 (0.303)	−0.209 (0.984)	−0.009 (0.137)	0.201 (1.241)
H6b	DIS → ATT	−0.028 (0.517)		0.022 (0.219)	0.009 (0.124)	0.013 (0.11)	−0.012 (0.136)	0.012 (0.126)	0.024 (0.186)	0.019 (0.234)	−0.131 (1.449)	0.15 (1.221)	0.04 (0.159)	−0.038 (0.581)	0.078 (0.425)	0.094 (0.458)	−0.002 (0.026)	0.095 (0.583)
H7	ATT → AI	0.662*** (12.885)	0.57*** (9.533)	0.741*** (12.938)	0.726*** (17.806)	0.016 (0.233)	0.683*** (15.461)	0.820*** (20.875)	0.138* (2.271)	0.74*** (16.797)	0.722*** (12.939)	0.018 (0.255)	0.743*** (11.187)	0.725*** (18.105)	0.018 (0.189)	0.801*** (13.039)	0.705*** (17.231)	0.096 (1.143)
H8	ATT → ENG	0.645*** (12.505)	0.786*** (32.221)	0.792*** (19.118)	0.786*** (30.533)	0.006 (0.125)	0.747*** (22.336)	0.822*** (23.899)	0.075 (1.558)	0.767*** (24.768)	0.813*** (23.392)	0.046 (0.989)	0.769*** (16.748)	0.79*** (29.575)	0.02 (0.321)	0.885*** (30.077)	0.747*** (26.056)	0.138* (2.445)

***p < 0.001.
**p < 0.01.
*p < 0.05.
ˆ p < 0.1.

Abbreviation: ATT- Attitude, HI= Haptic imagery, CON=Convenience, TA = Trend affinity, RA = Relative advantage, PC=Privacy concern, FC= Financial concern, PPR= Perceived performance risk, RM = Risk of malfunction, DIS = Discomfort, INS= Insecurity, OPT= Optimism, IN= Innovativeness.

significant challenge in digital adoption (Natarajan et al., 2024), did not significantly impact adoption attitudes, suggesting that younger consumers prioritize immersive digital experiences over security concerns. This finding aligns with studies on privacy trade-offs in digital environments, where digital convenience often outweighs security risks (Shahzad et al., 2024). Similarly, financial concerns did not emerge as a significant deterrent, despite expectations that high costs of VR hardware and digital assets might hinder adoption. This finding supports previous research indicating that younger consumers prioritize experience-driven spending over financial caution (Chakraborty et al., 2024). These insights highlight that while barriers exist, they are not strong enough to overshadow perceived benefits, reinforcing BRT's dual-process model, where positive reasoning exerts a more significant influence than negative concerns (Westaby, 2005).

The results strongly validate the BRT-TRI model, demonstrating that positive reasoning significantly influences metaverse adoption, while negative reasoning has a comparatively weaker impact. The findings confirm that positive attitudes (H7) drive adoption intentions (H8) and engagement (H9), reinforcing prior research on technology immersion and digital consumer behavior (Mehrotra et al., 2024). Additionally, this study highlights consumer engagement as a critical behavioral outcome, extending previous studies on experiential technology adoption (Nadeem et al., 2015). The results further suggest that metaverse adoption extends beyond initial novelty, influencing long-term digital interaction behaviors aligning with research on social presence and digital identity formation in virtual spaces (Yu, 2024). These insights support the dual-process model of BRT, affirming that consumers are more influenced by positive motivators than by adoption barriers (Westaby, 2005). By integrating psychological and technology-readiness perspectives, this study strengthens the understanding of metaverse adoption dynamics, contributing to broader discussions on digital consumer engagement and emerging technologies.

6.3. Theoretical implications

Several investigations have already been carried out to understand consumers' usage on the metaverse platforms (Adhini and Prasad, 2024; Yu, 2024; Ahn et al., 2024; Natarajan et al., 2024; Chakraborty et al., 2023; Xi et al., 2024; Taylor et al., 2024) but no single of them have addressed consumers resistance to it (Dwivedi et al., 2022; Gao et al., 2024; Gupta et al., 2024). Prior studies have also raised a necessity to adopt the holistic view to understand the facilitators and disruptors for adopting and using the metaverse technologies (Gao et al., 2024; Shukla et al., 2024; Sahu et al., 2020). Thus, to fill the gap in the extant literature, the present study implements BRT to understand the behaviors of metaverse users and enriches the theory's applicability by extending it to the context of novel technology. As a new and rapidly evolving field, BRT captures the dynamic aspects of user behavior well and makes original theoretical contributions to adapting behavioral frameworks to metaverse contexts. The results of this research are significant from a theoretical perspective and contribute to the existing literature in three key theoretical dimensions.

First, this study constructs and validates a framework rooted in BRT to explore factors impacting metaverse technology adoption, examining both 'reasons-for' and 'reasons-against' influences. Integrating drivers and deterrents into a unified model fills a significant literature gap, shedding light on how these factors shape user adoption intentions. The mixed-method approach identifies key drivers like haptic imagery, trend affinity, convenience, and relative advantage while highlighting deterrents such as privacy concerns, financial issues, and risks of malfunction. Notably, haptic imagery boosts consumer engagement and differentiation. At the same time, the research confirms privacy concerns and risk of malfunction as the most significant deterrents to user adoption, with financial concerns and perceived performance risks playing a more deficient role. These findings align with existing studies indicating heightened apprehension about privacy and unknown consequences of new technologies (Acikgoz et al., 2024; Zhang et al., 2019; Man et al., 2020; Waung et al., 2021). Moreover, contrary to industry reports highlighting high entry costs as a barrier, our findings show financial concerns were statistically insignificant among Generation Z and Millennials, who value novel experiences over costs (Dwivedi et al., 2023). While consumers view expenses like subscriptions or hardware as manageable, brands face significant platform development and maintenance costs.

Additionally, 46 % of marketers struggle with measuring ROI on metaverse investments (Chokshi, 2022). Implementing freemium consumer models (Kumar et al., 2024) and clear ROI metrics for brands can help mitigate financial barriers and drive adoption. The factors categorized as 'reasons for' and 'reasons against' within our framework, grounded in BRT, have received limited attention in the context of the metaverse. Consequently, these factors substantiate our model, demonstrating a notable impact of both types of reasons on user attitudes. This framework enhances current academic discourse and offers a detailed insight into how users interact with metaverse technology.

Secondly, by making technology readiness a value and a separate variable in the BRT model, we build on previous research on what affects the adoption of the metaverse. This paper investigates the link between values and the metaverse, contributing valuable insights into how values affect the readiness to use such technologies. This extension of the BRT framework identifies technology readiness compatibility as a critical factor influencing the adoption of new technology and its usage over time.

Third, a systematic literature review by Sahu et al. (2020) points to the need to develop BRT by adding new variables, as minimal studies have investigated new dependent variables within the framework of BRT. Based on this call, Claudy et al. (2015) and Gao et al. (2024) agree that future research should be conducted in emerging areas. This study answers that call by empirically showing the relationship between metaverse-related attitudes and their effects on engagement, extending the BRT framework to this context. Our findings show that a positive attitude towards the metaverse increases adoption intention and engagement. While prior research has examined the attitude–intention behavior gap, this paper further contributes to the literature by highlighting the role of engagement. Users are more likely to exhibit consistent usage behavior and actively engage with the virtual environment (Shahzad et al., 2024), expanding our knowledge of the psychological aspects of technology adoption. This study has begun to develop theoretical concepts, forming the basis for future work to build and elaborate upon.

6.4. Practical implications

The metaverse offers brands opportunities to engage with customers differently than before. The present study postulates that technology readiness, which is comprised of favorable factors such as optimism and innovativeness and unfavorable factors such as discomfort and security, influences users' attitudes toward the metaverse. The outcomes of the study show that users' perception of imagery convenience, trend affinity, and relative advantage is predominantly influenced by optimism and innovativeness (Parasuraman and Colby, 2015; Blut and Wang, 2020). This can help evoke emotions and alleviate worries related to security risks (Jan et al., 2023; Sahu et al., 2020).

Our study shows that worries about privacy and financial concerns, as well as perceived performance risks and risks of malfunction, are the key obstacles to embracing the metaverse. Technology and robust data encryption methods can enhance privacy safeguards and data handling guidelines (Zhao et al., 2023a,b; Vimalkumar et al., 2021). It is crucial to clarify how the metaverse safeguards data and handles third-party data sharing while empowering users to manage their details to establish trust (as mentioned by Dwivedi et al., 2022; Roy et al., 2020). To address performance challenges in this context, developers need to guarantee the presence of failover mechanisms, including automated setups with

real-time monitoring and maintenance alongside predictive measures for potential failures. Conducting assessments amid demand can also enhance users' trust (Waung et al., 2021; Hadi et al., 2024). With the rise of surveillance capitalism and the growing importance of third-party data in settings, companies need to tackle these challenges by being open about their practices, adhering to a rising array of privacy regulations, and empowering individuals to have a say in how their data is utilized. A crucial parameter for the security consideration will be establishing a user-friendly privacy interface that enables users to safeguard and manage their data choices and ultimately increases security.

The findings suggest that factors like haptic imagery, trend affinity, relative advantage, and convenience drive the users' metaverse adoption. Haptic images significantly influence the users' engagement due to their realism features, which provide more authentic brand-related interactions. Thus, marketers can capitalize on opportunities, like 3D experiences and virtual touch simulations, to enhance engagement levels (Silva et al., 2021). Additionally, convenience plays a role in user adoption as people expect quick access to services. Making it easier for people to get into it all smoothly and comfortably when brands focus more on creating a user interface with navigation and convenient ways to use metaverse apps will help break down barriers to adoption. Furthermore, the trend factor plays a role in motivating folks – this means that including influencer marketing strategies and offering collectibles as well as exclusive virtual experiences that are limited edition could catch the attention of Generation Z and Millennials, who are all about embracing the latest tech trends (Kaur et al., 2024; Arya et al., 2024).

The results of this research can have implications for different sectors. For example, in education, haptic imagery and the incorporation of intelligence enhance online classrooms to make them engaging and efficient (as shown by Sagnier et al., 2020). Moreover, the metaverse could benefit the healthcare industry by offering telemedicine, virtual physiotherapy, environmental therapy, and AI-supported diagnosis services that enhance healthcare accessibility and reduce costs (Zhang et al., 2025). Taking inspiration from companies such as Gucci and Walmart, who have effectively built brand environments in virtual reality settings, can pave the way for marketers to craft strategies for retail and advertising purposes in the future. These future techniques should use AI-oriented customization technologies to devise unique shopping experiences, such as virtual product testings and trial rooms exclusively designed for the metaverse technology.

A key takeaway from the study is the role of engagement as an outcome of metaverse adoption. However, a deeper understanding of engagement is necessary. Our findings suggest that engagement should be viewed along a spectrum, ranging from passive engagement, such as observing or browsing virtual spaces, to active engagement, which includes social interactions, gamified experiences, and co-creation activities (Roy et al., 2020; Arya et al., 2024). Businesses should focus on strategies encouraging active engagement, such as integrating gamification elements, rewarding user participation with digital incentives, and fostering social connectivity through virtual events. AI-driven personalization, interactive avatars, and immersive branded spaces can enhance user involvement and sustain long-term engagement.

The findings underscore the importance of segment-specific strategies in driving metaverse adoption. Regular users, already familiar with immersive digital environments, benefit from advanced features such as enhanced interactivity, personalization, and exclusive content to sustain engagement. In contrast, non-regular users may require guided onboarding experiences, educational initiatives, and simplified interfaces to ease adoption barriers. Income disparities indicate that flexible pricing models, including freemium access, microtransactions, and tiered subscriptions, can help address financial concerns among lower-income consumers while offering premium experiences to high-income users. Differences in education levels suggest marketers should provide tailored communication strategies, with more technical and in-depth content for highly educated users and user-friendly, step-by-step guides for less tech-savvy consumers. Generational variations further highlight the need for differentiated engagement strategies—Gen Z, being digital natives, may respond well to gamified experiences, social interactions, and influencer-driven promotions.

In contrast, Millennials may be more influenced by tangible benefits such as professional networking and investment opportunities within the metaverse. The results also suggest that marketing strategies emphasizing trend-based appeal may be more effective for male consumers. In contrast, strategies targeting female users should focus on alternative motivational drivers beyond trend affinity to enhance adoption intention. Additionally, businesses should consider financial incentives, such as subscription flexibility or installment-based purchases, to mitigate financial concerns among male users while ensuring platform reliability and performance stability to address their risk perceptions.

6.5. Limitations and future research directions

Although the current study offers nuanced insights on metaverse adoption, it is essential to acknowledge the shortcomings that could be addressed in future research. To begin with, the first phase, the qualitative study, was solely focused on the semi-structured interviews with the developers and marketers of the metaverse. While this phase provided valuable industry-related viewpoints, they might be biased by the samples and unable to capture end users' views. Future studies could resolve this issue by considering a larger spectrum of stakeholders, such as consumers, platform developers, and policymakers, to understand the holistic perspectives of the metaverse adoption intention. Furthermore, future research can incorporate a longitudinal approach to monitor the evolving customer's attitudes and actions over time to enhance the generalizability of the qualitative study findings.

Second, the quantitative phase primarily consisted of responses from Gen Z and Millennials, who have extensive experience with virtual environments such as AltspaceVR, Decentraland, and other blockchain-based platforms. While this focus aligns with the most active metaverse user base, it introduces sample bias by potentially excluding perspectives from older generations or less tech-savvy users. Future studies could expand the sample to include users across different demographics and technology adoption levels, ensuring greater representativeness.

Third, self-reporting bias is an inherent limitation of survey-based research, as participants may provide socially desirable responses or misinterpret specific questions. To mitigate this concern, future research could incorporate behavioral data (e.g., metaverse usage logs, eye-tracking data, or transaction records) alongside self-reported measures to enhance the objectivity and reliability of findings.

Fourth, while privacy concerns emerged as a significant barrier to metaverse adoption in our study, we acknowledge that our analysis does not comprehensively examine the broader data security landscape within the metaverse. Critical issues such as surveillance capitalism, third-party data sharing, and decentralized governance require further exploration. Future research could delve deeper into these nuances to provide a more robust understanding of how privacy risks influence metaverse adoption decisions. A detailed examination of regulatory frameworks and consumer attitudes toward privacy protection mechanisms could further strengthen this area of inquiry.

Fifth, this study employs a cross-sectional research design, which limits the ability to establish causal relationships. Future research could benefit from longitudinal or experimental designs to validate and expand upon our findings. Additionally, post-adoptive behavior remains an important area for exploration. While our study primarily focuses on initial adoption intent, future research could examine user retention, sustained engagement, and the long-term impact of metaverse experiences. Furthermore, while we introduce engagement as an outcome variable, we acknowledge that our study does not differentiate between passive and active engagement or analyze how different forms of

engagement shape long-term consumer behavior. Future research could explore these distinctions, assessing whether different engagement types (e.g., transactional engagement vs. immersive participation) lead to varying adoption patterns and brand loyalty effects. Sixth, while our study successfully integrates qualitative and quantitative data using a mixed-methods approach, we recognize the inherent challenges in integrating insights from both phases. Differences in interpretation between expert opinions (qualitative findings) and user behavior (quantitative results) may introduce limitations in how conclusions are drawn. Future studies could explore more advanced methodological triangulation techniques to strengthen the integration of qualitative and quantitative insights.

Finally, our model is primarily grounded in BRT, which we expand by including technology readiness and engagement as key factors affecting adoption. Future studies could build upon our framework by integrating additional constructs from other behavioral theories, such as Expectation-Confirmation Theory (ECT) or the Extended Unified Theory of Acceptance and Use of Technology (UTAUT2), to further enhance the understanding of metaverse adoption. While there are limitations, this study lays a strong foundation for future research by identifying the main drivers and obstacles to metaverse adoption. By recognizing these limitations, we aim to inspire further investigation and methodological improvements to strengthen the depth and relevance of metaverse adoption research.

6.6. Conclusion

This research study thoroughly investigates consumer metaverse adoption by integrating the BRT and the TRI to identify the motivators and inhibitors that significantly influence attitudes, adoption, and engagement. Integrating a mixed-methods approach, combining qualitative insights from industry experts and quantitative insights validated by extensive surveys, this study strengthens the theoretical and practical application of how consumers perceive and interact in the virtual world. The findings suggest that relative advantage, haptic imagery, and convenience positively impact the metaverse adoption intention, which validates the importance of the positive enablers in shaping consumer attitudes. Contrary to that, barriers such as financial and privacy concerns exert a relatively weaker influence, reinforcing the BRT premises that the positive influence prevails over the negative impact in the decision-making process. Furthermore, the present study's findings also highlight that attitude has emerged as the determining factor leading to adoption intentions and engagement; hence, metaverse marketers and policymakers should focus on enhancing consumers' perceptions and initiating more active participation in the metaverse. The study makes several theoretical contributions by extending BRT into the metaverse adoption context, integrating technology readiness as a critical factor, and distinguishing between motivators and barriers within digital ecosystems. The managerial implications emphasize the need for personalized, gamified, and interactive metaverse experiences to enhance adoption while addressing residual trust and security concerns through transparent policies and enhanced user protections. While this research provides valuable insights, limitations remain. The study focuses on Gen Z and Millennial consumers, limiting generalizability to older demographics or emerging user groups. Future research could explore longitudinal shifts in adoption patterns and comparative analyses across different metaverse applications, such as gaming, education, and retail. In conclusion, this study contributes to the growing discourse on digital consumer behavior, offering a structured, theory-driven approach to understanding metaverse adoption and engagement. As the metaverse evolves, businesses, policymakers, and researchers must adapt to emerging trends, user expectations, and technological advancements to foster a sustainable and inclusive digital ecosystem.

CRediT authorship contribution statement

Sigma Soni: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Parvinder Arora:** Writing – original draft, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization, Writing – review & editing. **Dharun Kasilingam:** Writing – review & editing, Software, Methodology, Formal analysis, Data curation, Funding acquisition. **Varsha Jain:** Writing – review & editing.

Funding information

This study was supported by the Indian Institute of Management Kozhikode under the Large Grant Research Project (LGRP) grant ID LGRP/2023–24/04. The authors sincerely appreciate the institute's financial assistance and research support, which played a crucial role in facilitating this study.

Declaration of competing interest

The authors declare that they have no known competing interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix 1. A Discussion Guide

Research Objective- To understand the metaverse use by the industry

Stage 1- Perspective on Metaverse

1. What are your views on the metaverse? What is the first thing that comes to mind when you think of the metaverse?
2. Would you please share your experience using technology such as Metaverse?
3. [Promts: Discuss their first experience while using new technology. What stands out for them when using and adapting/accessing such technology? Did somebody or a competition influence them to use it? Or are they proclaimed early users of the technology?
4. Can you please elaborate on what worked well and what did not about the metaverse?
5. [Promts: For example, was it the customer demand or competition from the rival that motivated you to use the technology? Was it the quality it offered? Was it its ability to easily access and stimulate customers' decision-making?]
6. Would you please discuss any strategies or features you didn't like about the metaverse?
7. [Prompts: For example, was it possible to access it? Did you have other concerns while accessing the technology (not user-friendly? Cost? Customer awareness?) Is there anything else you would like to replace with metaverse? If so, why?]
8. Can you give us a brief idea of the different industries adopting metaverse in practice?

Stage 2- Industry Specific Question

1. To what extent would you think the evolving technology, such as metaverse, will be helpful to the category your company is in, and how?
2. How would meta help this industry?
3. What are the reasons behind marketers' hesitation to adopt technological innovations?

Stage 3- Responses and Outcome from Metaverse

1. How would you describe your experience after receiving feedback from the users who have tried new technology (metaverse)

[Prompts: For example, have they done a trial on customers? Did it affect their experience positively/negatively? What are the reasons the industry should consider before adopting the metaverse? Also, can you describe any situations where you would like to take different steps in managing your company by offering metaverse experiences to the customer].

Stage 4- Strategies Of Meta

1. To what extent would businesses accept new technology such as metaverse? Can you give us a few reasons to consider investing in the metaverse to expand the industry?
2. As a practitioner, what do you think about adopting technology such as metaverse, which is helpful for consumers?
3. From your perspective, what are the critical consumer preferences that will change with the adoption of metaverse into the industry?
4. To what extent would you like to prefer/not to pick the adoption of a metaverse in your business strategy? Why?
5. From your perspective, what results can the industry see as a whole after adopting metaverse technology?
6. Also, what reforms are needed to improve before adopting metaverse technology?

Stage 5- Future Of Meta from Experts

1. What do companies expect from the Meta, users, and platforms?
2. What will be the future of Meta?
3. How will it evolve and emerge across the industry?

Appendix 2. Respondents' demography for the Phase I- Qualitative Study

Code No	Age group	Gender	Education level	Designation	Organisation	Interview Duration (in mins)
1	18–25	Male	Post Graduate	Digital Marketing Manager	E-commerce Firm A	37
2	26–33	Male	Graduate	Tech development Lead	Technology Company A	43
3	18–25	Female	Graduate	UI/UX Developer	Design Agency A	45
4	34–41	Female	Post Graduate	Digital Marketing Manager	Retail Chain A	39
5	42–49	Male	Post Graduate	Digital Marketing Manager	Fashion Company A	50
6	26–33	Male	Post Graduate	Digital Marketing Manager	Gaming Platform A	58
7	18–25	Male	Post Graduate	Digital Marketing Manager	Media and Tech Consulting Firm A	70
8	34–41	Female	Post Graduate	Tech development Lead	Entertainment Company A	53
9	26–33	Male	Graduate	UI/UX Developer	Financial Services Firm A	37
10	18–25	Male	Graduate	UI/UX Developer	Automotive Company A	64
11	18–25	Female	Post Graduate	Tech development Lead	Technology Company B	39
12	34–41	Male	Post Graduate	Digital Innovation & Media Manager	Retail Chain B	54
13	34–41	Female	Post Graduate	Tech development Lead	Real Estate Firm A	43
14	42–49	Male	Graduate	Tech development Lead	Consulting Firm B	34
15	26–33	Male	Graduate	Tech development Lead	Fashion Company A	55
16	26–33	Male	Post Graduate	Digital Innovation & Media Manager	Digital Fashion platform A	38
17	34–41	Female	Post Graduate	Digital Marketing Manager	Logistics Company A	35
18	42–49	Male	Post Graduate	Digital Innovation & Media Manager	Tourism Board A	43
19	26–33	Female	Post Graduate	Digital Marketing Manager	Technology Company A	63
20	34–41	Female	Post Graduate	Digital Marketing Manager	Media Company A	68
21	42–49	Male	Post Graduate	Tech development Lead	Government Technology A	51
22	42–49	Male	Post Graduate	Digital Innovation & Media Manager	Retail Chain B	55
23	26–33	Male	Graduate	UI/UX Developer	Entertainment Company B	38
24	34–41	Female	Post Graduate	Digital Innovation & Media Manager	Gaming Platform B	68
25	18–25	Male	Post Graduate	Digital Marketing Manager	Media and Tech Consulting firm B	44
26	34–41	Male	Post Graduate	Digital Marketing Manager	Automotive Company B	69
27	18–25	Male	Post Graduate	Digital Marketing Manager	Gaming Platform B	63
28	34–41	Female	Post Graduate	Digital Marketing Manager	Gaming Platform B	44
29	26–33	Female	Graduate	Tech development Lead	Retail Chain B	43
30	34–41	Female	Post Graduate	Strategy Manager	Metaverse Agency A	39
31	18–25	Male	Post Graduate	Strategy Manager	Metaverse Agency A	46
32	18–25	Male	Post Graduate	Digital Innovation & Media Manager	Technology Company A	58
33	34–41	Male	Post Graduate	Digital Innovation & Media Manager	Entertainment Company A	76
34	26–33	Female	Graduate	Tech development Lead	Advertising Agency A	39
35	34–41	Male	Graduate	Tech development Lead	EdTech Company A	41
36	34–41	Female	Graduate	UI/UX Developer	Marketing Agency B	54
37	26–33	Male	Post Graduate	Digital Innovation & Media Manager	Tourism Board A	43

(*continued on next page*)

(*continued*)

Code No	Age group	Gender	Education level	Designation	Organisation	Interview Duration (in mins)
38	26–33	Male	Graduate	UI/UX Developer	EdTech Company A	34
39	42–49	Male	Post Graduate	Digital Marketing Manager	Advertising Agency A	55
40	26–33	Male	Graduate	UI/UX Developer	Marketing Agency A	38
41	42–49	Male	Post Graduate	Tech development Lead	Association for Sustainable Fashion	44
42	>50	Male	Post Graduate	Digital Marketing Manager	Marketing Agency A	69
43	42–49	Female	Post Graduate	Digital Innovation & Media Manager	Financial Services Firm B	63
44	>50	Female	Graduate	UI/UX Developer	EdTech Company A	40
45	34–41	Male	Post Graduate	Tech development Lead	Advertising Agency A	43
46	26–33	Male	Graduate	UI/UX Developer	Gaming Platform B	46
47	26–33	Female	Post Graduate	Digital Marketing Manager	E-commerce Firm A	67
48	26–33	Male	Post Graduate	Digital Marketing Manager	Marketing Agency B	56

Data availability

Data will be made available on request.

References

Accenture, 2023. Growing consumer and business interest in the Metaverse is expected to fuel a trillion-dollar opportunity for commerce. Accenture finds. https://newsroom.accenture.com/. https://newsroom.accenture.com/news/2023/growing-consumer-and-business-interest-in-the-metaverse-expected-to-fuel-trillion-dollar-opportunity-for-commerce-accenture-finds.

Acikgoz, F., Borulu, B., Bölen, M.C., 2024. How does obsolescence risk influence consumer resistance to smartwatches? Inf. Technol. People. https://doi.org/10.1108/itp-08-2023-0843.

Adhini, N.V., Prasad, C.V., 2024. Perceptions and drivers of the metaverse adoption: a mixed-methods study. Int. J. Consum. Stud. 48 (4), e13069. https://doi.org/10.1111/ijcs.13069.

Ahn, S., Jin, B.E., Seo, H., 2024. Why do people interact and buy in the Metaverse? Self-Expansion perspectives and the impact of hedonic adaptation. J. Bus. Res. 175, 114557. https://doi.org/10.1016/j.jbusres.2024.114557.

Ajzen, I., 1991. The theory of planned behavior. Organ. Behav. Hum. Decis. Process. 50 (2), 179–211. https://doi.org/10.1016/0749-5978(91)90020-T.

Akour, I.A., Al-Maroof, R.S., Alfaisal, R., Salloum, S.A., 2022. A conceptual framework for determining metaverse adoption in higher institutions of gulf area: an empirical study using hybrid SEM-ANN approach. Comput. Educ.: Artif. Intell. 3, 100052. https://doi.org/10.1016/j.caeai.2022.100052.

Alexander, B., Blazquez, M., Chrimes, C., Boardman, R., 2024. The role of immersive spaces on the customer experience: an exploration of fashion metaverses. Psychol. Market. https://doi.org/10.1002/mar.22140.

Armstrong, J.S., Overton, T.S., 1977. Estimating nonresponse bias in mail surveys. J. Mark. Res. 14 (3), 396–402. https://doi.org/10.1177/002224377701400320.

Arya, V., Sambyal, R., Sharma, A., Dwivedi, Y.K., 2024. Brands are calling your AVATAR in Metaverse–A study to explore XR-based gamification marketing activities & consumer-based brand equity in virtual world. J. Consum. Behav. 23 (2), 556–585. https://doi.org/10.1002/cb.2214.

Auxier, Arbanas. Young people are already there while we wait for the metaverse to materialize (2023, July 27). Delloite Insights. https://www2.deloitte.com/us/en/insights/industry/technology/gen-z-and-millennials-are-metaverse-early-adopters.html.

Bentler, P.M., Bonett, D.G., 1980. Significance tests and goodness of fit in the analysis of covariance structures. Psychol. Bull. 88 (3), 588–606. https://doi.org/10.1037/0033-2909.88.3.588.

Bilgihan, A., Leong, A.M., Okumus, F., Bai, J., 2024. Proposing a metaverse engagement model for brand development. J. Retailing Consum. Serv. 78, 103781. https://doi.org/10.1016/j.jretconser.2024.103781.

Bezbaruah, S., Trivedi, J., 2020. Branded content: a bridge building Gen Z's consumer–brand relationship. Vision. J. Business Perspect. 24 (3), 300–309. https://doi.org/10.1177/0972262920930167.

Blut, M., Wang, C., 2020. Technology readiness: a meta-analysis of conceptualizations of the construct and its impact on technology usage. J. Acad. Market. Sci. 48 (4), 649–669. https://doi.org/10.1007/s11747-019-00680-8.

Brack, A., Benkenstein, M., 2014. Responses to other similar customers in a service setting – analyzing the moderating role of perceived performance risk. J. Serv. Market. 28 (2), 138–146. https://doi.org/10.1108/jsm-05-2012-0089.

Brakus, J.J., Schmitt, B.H., Zarantonello, L., 2009. Brand experience: what is it? How is it measured? Does it affect loyalty? J. Market. 73 (3), 52–68. https://doi.org/10.1509/jmkg.73.3.052.

Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. Qual. Res. Psychol. 3 (2), 77–101.

Buhalis, D., Leung, D., Lin, M., 2023. Metaverse is a disruptive technology revolutionizing tourism management and marketing. Tour. Manag. 97, 104724. https://doi.org/10.1016/j.tourman.2023.104724.

Chakraborty, D., Patre, S., Tiwari, D., 2023. Metaverse mingle: discovering dating intentions in metaverse. J. Retailing Consum. Serv. 75, 103509. https://doi.org/10.1016/j.jretconser.2023.103509.

Chakraborty, D., Polisetty, A., Rana, N.P., 2024. Consumers' continuance intention towards metaverse-based virtual stores: a multi-study perspective. Technol. Forecast. Soc. Change 203, 123405. https://doi.org/10.1016/j.techfore.2024.123405.

Chekembayeva, G., Garaus, M., Schmidt, O., 2023. The role of time convenience and (anticipated) emotions in AR mobile retailing application adoption. J. Retailing Consum. Serv. 72, 103260. https://doi.org/10.1016/j.jretconser.2023.103260.

Chen, G., Bliese, P.D., Mathieu, J.E., 2005. Conceptual framework and statistical procedures for delineating and testing multilevel theories of homology. Organ. Res. Methods 8 (4), 375–409. https://doi.org/10.1177/1094428105280056.

Chin, W.W., Thatcher, J.B., Wright, R.T., Steel, D., 2013. Controlling for common method variance in PLS analysis: the measured latent marker variable approach. New Perspectives in Partial Least Squares and Related Methods, pp. 231–239. https://doi.org/10.1007/978-1-4614-8283-3_7.

Choi, H., Park, J., Jung, Y., 2018. The role of privacy fatigue in online privacy behavior. Comput. Hum. Behav. 81, 42–51. https://doi.org/10.1016/j.chb.2017.12.001.

Chokshi, A., 2022. Forrester study finds 228 percent ROI when modernizing applications on Azure PaaS. Retrieved from. https://azure.microsoft.com/en-us/blog/forrester-study-finds-228-percent-roi-when-modernizing-applications-on-azure-paas/.

Claudy, M.C., Peterson, M., 2014. Understanding the underutilization of urban bicycle commuting: a behavioral reasoning perspective. J. Publ. Pol. Market. 33 (2), 173–187. https://doi.org/10.1509/jppm.13.087.

Claudy, M.C., Peterson, M., O'driscoll, A., 2013. Understanding the attitude-behavior gap for renewable energy systems using behavioral reasoning theory. J. Macromarketing 33 (4), 273–287. https://doi.org/10.1177/0276146713481605.

Claudy, M.C., Garcia, R., O'Driscoll, A., 2015. Consumer resistance to innovation—a behavioral reasoning perspective. J. Acad. Market. Sci. 43, 528–544. https://doi.org/10.1007/s11747-014-0399-0.

Copeland, M.T., 1923. Relation of consumers' buying habits to marketing methods. Harv. Bus. Rev. 1 (2), 282–289.

Cresswell, J., 2013. Qualitative Inquiry & Research Design: Choosing Among Five Approaches.

Creswell, J.W., Clark, V.L.P., 2017. Designing and Conducting Mixed Methods Research. Sage publications.

Cruz-Cárdenas, J., Guadalupe-Lanas, J., Ramos-Galarza, C., Palacio-Fierro, A., 2021. Drivers of technology readiness and motivations for consumption in explaining the tendency of consumers to use technology-based services. J. Bus. Res. 122, 217–225. https://doi.org/10.1016/j.jbusres.2020.08.054.

Davis, F.D., 1989. Technology acceptance model: TAM. Al-Suqri, MN, Al-Aufi, AS: Informat. Seek. Behav. Technol. Adopt. 205, 219. https://quod.lib.umich.edu/b/busadwp/images/b/1/4/b1409190.0001.001.pdf.

Delgosha, M.S., Hajiheydari, N., 2020. On-demand service platforms pro/anti adoption cognition: examining the context-specific reasons. J. Bus. Res. 121, 180–194. https://doi.org/10.1016/j.jbusres.2020.08.031.

Dhir, A., Koshta, N., Goyal, R.K., Sakashita, M., Almotairi, M., 2021. Behavioral reasoning theory (BRT) perspectives on E-waste recycling and management. J. Clean. Prod. 280, 124269. https://doi.org/10.1016/j.jclepro.2020.124269.

Dwivedi, Y.K., Balakrishnan, J., Mishra, A., De Bock, K.W., Al-Busaidi, A.S., 2024. The role of embodiment, experience, and self-image expression in creating continuance intention in the metaverse. Technol. Forecast. Soc. Change 203, 123402. https://doi.org/10.1016/j.techfore.2024.123402.

Dwivedi, Y.K., Hughes, L., Baabdullah, A.M., Ribeiro-Navarrete, S., Giannakis, M., Al-Debei, M.M., et al., 2022. Metaverse beyond the hype: multidisciplinary perspectives on emerging challenges, opportunities, and agenda for research, practice and policy. Int. J. Inf. Manag. 66, 102542. https://www.sciencedirect.com/science/article/pii/S0268401222000767.

Dwivedi, Y.K., Hughes, L., Wang, Y., Alalwan, A.A., Ahn, S.J., Balakrishnan, J., et al., 2023. Metaverse marketing: how the metaverse will shape the future of consumer research and practice. Psychol. Market. 40 (4), 750–776. https://doi.org/10.1002/mar.21767.

Eggenschwiler, M., Linzmajer, M., Roggeveen, A.L., Rudolph, T., 2024. Retailing in the metaverse: a framework of managerial considerations for success. J. Retailing Consum. Serv. 79, 103791. https://doi.org/10.1016/j.jretconser.2024.103791.

Engel, J., Blackwell, R., Miniard, P., 2001. Consumer Behavior, ninth ed. Dryden Press.

Erensoy, A., Mathrani, A., Schnack, A., Elms, J., Baghaei, N., 2024. Consumer behavior in immersive virtual reality retail environments: a systematic literature review using the stimuli-organisms-responses (S-O-R) model. J. Consum. Behav. https://doi.org/10.1002/cb.2374.

Erskine, M.A., Khojah, M., McDaniel, A.E., 2019. Location selection using heat maps: relative advantage, task-technology fit, and decision-making performance. Comput. Hum. Behav. 101, 151–162. https://doi.org/10.1016/j.chb.2019.07.014.

Featherman, M., Jia, S.J., Califf, C.B., Hajli, N., 2021. The impact of new technologies on consumers beliefs: reducing the perceived risks of electric vehicle adoption. Technol. Forecast. Soc. Change 169, 120847. https://www.sciencedirect.com/science/article/pii/S0040162521002791.

Fishbein, M., Ajzen, 1975. Attitude, Intention, and Behaviour: an Introduction to Theory and Research. Addison-Wesley, Reading. MA.

Flavián, C., Ibáñez-Sánchez, S., Orús, C., 2019. The impact of virtual, augmented and mixed reality technologies on the customer experience. J. Bus. Res. 100, 547–560. https://doi.org/10.1016/j.jbusres.2018.10.050.

Gartner, 2022. Gartner predicts 25% of people will spend at least one hour per day in the Metaverse by 2026. Retrieved October 2024, from. https://www.gartner.com/en/newsroom/press-releases/2022-02-07-gartner-predicts-25-percent-of-people-will-spend-at-least-one-hour-per-day-in-the-metaverse-by-2026.

Gaskell, G., 2000. Individual and Group Interviewing. *Qualitative Researching with Text, Image and Sound*, pp. 38–56.

Gao, H., Chong, A.Y.L., Bao, H., 2024. Metaverse: literature review, synthesis and future research agenda. J. Comput. Inf. Syst. 64 (4), 533–553. https://doi.org/10.1080/08874417.2023.2233455.

Ghali, Z., Rather, R.A., Khan, I., 2024. Investigating metaverse marketing-enabled consumers' social presence, attachment, engagement and (re) visit intentions. J. Retailing Consum. Serv. 77, 103671. https://www.sciencedirect.com/science/article/pii/S0969698923004228.

GSMA, 2022. Exploring the metaverse and digital future. GSMA Assoc. https://www.gsma.com/asia-pacific/wp-content/uploads/2022/02/270222-Exploring-the-metavere-and-the-digital-future.pdf.

Gupta, A., Arora, N., 2017. Understanding determinants and barriers of mobile shopping adoption using behavioral reasoning theory. J. Retailing Consum. Serv. 36, 1–7. https://doi.org/10.1016/j.jretconser.2016.12.012.

Gupta, R., Rathore, B., Biswas, B., Jaiswal, M., Singh, R.K., 2024. Are we ready for metaverse adoption in the service industry? Theoretically exploring the barriers to successful adoption. J. Retailing Consum. Serv. 79, 103882. https://doi.org/10.1016/j.jretconser.2024.103882.

Hadi, R., Melumad, S., Park, E.S., 2024. The Metaverse: a new digital frontier for consumer behavior. J. Consum. Psychol. 34 (1), 142–166. https://doi.org/10.1002/jcpy.1356.

Hair, J.F., Hult, G.T.M., Ringle, C.M., Sarstedt, M., 2017. A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM), second ed. Sage.

Hair, J.F., Sarstedt, M., Ringle, C.M., Gudergan, S.P., 2019. Advanced Issues in Partial Least Squares Structural Equation Modeling (PLS-SEM). Sage.

Hennig-Thurau, T., Aliman, D.N., Herting, A.M., Cziehso, G.P., Linder, M., Kübler, R.V., 2023. Social interactions in the metaverse: framework, initial evidence, and research roadmap. J. Acad. Market. Sci. 51 (4), 889–913. https://doi.org/10.1007/s11747-022-00908-0.

Henseler, J., Ringle, C.M., Sarstedt, M., 2015. A new criterion for assessing discriminant validity in variance-based structural equation modeling. J. Acad. Market. Sci. 43, 115–135.

Henseler, J., Ringle, C.M., Sarstedt, M., 2016. Testing measurement invariance of composites using partial least squares. Int. Mark. Rev. 33 (3), 405–431.

Hilken, T., Chylinski, M., Keeling, D.I., Heller, J., de Ruyter, K., Mahr, D., 2022. How to strategically choose or combine augmented and virtual reality for improved online experiential retailing. Psychol. Market. 39 (3), 495–507. https://doi.org/10.1002/mar.21600.

Hollebeek, L.D., Glynn, M.S., Brodie, R.J., 2014. Consumer brand engagement in social media: Conceptualization, scale development and validation. J. Interact. Market. 28 (2), 149–165. https://doi.org/10.1016/j.intmar.2013.12.002.

Hollensen, S., Kotler, P., Opresnik, M.O., 2022. Metaverse–the new marketing universe. J. Bus. Strat. 44 (3), 119–125. https://doi.org/10.1108/JBS-01-2022-0014.

Hu, L.T., Bentler, P.M., 1999. Cutoff criteria for fit indexes in covariance structure analysis: conventional criteria versus new alternatives. Struct. Equ. Model.: A Multidiscip. J. 6 (1), 1–55. https://doi.org/10.1080/10705519909540118.

Huang, D., Jin, X., Huang, J., Kong, S., 2023. Tourist acceptance of robot chefs in gastronomy experiences: a behavioural reasoning perspective. Tour. Manag. Perspect. 48, 101172. https://doi.org/10.1016/j.tmp.2023.101172.

Huang, T.L., Liao, S.L., 2017. Creating e-shopping multisensory flow experience through augmented-reality interactive technology. Internet Res. 27 (2), 449–475. https://doi.org/10.1108/IntR-11-2015-0321.

Huang, Y., Qian, L., 2021. Understanding the potential adoption of autonomous vehicles in China: the perspective of behavioral reasoning theory. Psychol. Market. 38 (4), 669–690. https://doi.org/10.1002/mar.21465.

Hwang, J., Choe, J.Y., 2019. Exploring perceived risk in building successful drone food delivery services. Int. J. Contemp. Hospit. Manag. 31 (8), 3249–3269. https://doi.org/10.1108/IJCHM-07-2018-0558.

Ibáñez-Sánchez, S., Orús, C., Flavián, C., 2022. Augmented reality filters on social media. Analyzing the drivers of playability based on uses and gratifications theory. Psychol. Market. 39 (3), 559–578. https://doi.org/10.1002/mar.21639.

Inman, J.J., Nikolova, H., 2017. Shopper-facing retail technology: a retailer adoption decision framework incorporating shopper attitudes and privacy concerns. J. Retailing 93 (1), 7–28. https://doi.org/10.1016/j.jretai.2016.12.006.

Ivanov, A., Head, M., Biela, C., 2023. Mobile shopping decision comfort using augmented reality: the effects of perceived augmentation and haptic imagery. Asia Pac. J. Mark. Logist. 35 (8), 1917–1934. https://doi.org/10.1108/APJML-06-2022-0518.

Jafar, R.M.S., Ahmad, W., Sun, Y., 2023. Unfolding the impacts of metaverse aspects on telepresence, product knowledge, and purchase intentions in the metaverse stores. Technol. Soc. 74, 102265. https://doi.org/10.1016/j.techsoc.2023.102265.

Jan, I.U., Ji, S., Kim, C., 2023. What (de) motivates customers to use AI-powered conversational agents for shopping? The extended behavioral reasoning perspective. J. Retailing Consum. Serv. 75, 103440. https://doi.org/10.1016/j.jretconser.2023.103440.

Jung, I., Quan, W., Yu, J., Han, H., 2023. Are you ready for robot services? Exploring robot-service adoption behaviors of hotel-goers. Int. J. Hospit. Manag. 109, 103404. https://doi.org/10.1016/j.ijhm.2022.103404.

Kasilingam, D.L., 2020. Understanding the attitude and intention to use smartphone chatbots for shopping. Technol. Soc. 62, 101280. https://doi.org/10.1016/j.techsoc.2020.101280.

Kasilingam, D., Krishna, R., 2022. Understanding the adoption and willingness to pay for Internet of Things services. Int. J. Consum. Stud. 46 (1), 102–131. https://doi.org/10.1111/ijcs.12648.

Kaur, J., Mogaji, E., Paliwal, M., Jha, S., Agarwal, S., Mogaji, S.A., 2024. Consumer behavior in the metaverse. J. Consum. Behav. 23 (4), 1720–1738. https://doi.org/10.1002/cb.2298.

Kautish, P., Purohit, S., Filieri, R., Dwivedi, Y.K., 2023. Examining the role of consumer motivations to use voice assistants for fashion shopping: the mediating role of awe experience and eWOM. Technol. Forecast. Soc. Change 190, 122407. https://doi.org/10.1016/j.techfore.2023.122407.

Ki, C.W.C., Li, C., Chenn, A.S., Chong, S.M., Cho, E., 2024. Wise consumer choices in online secondhand luxury (OSHL) shopping: an integrated model of motivations, attitudes, and purchase intentions for OSHL as wise, conspicuous, and sustainable consumption. J. Retailing Consum. Serv. 76, 103571. https://doi.org/10.1016/j.jretconser.2023.103571.

Kim, J., 2021. Advertising in the metaverse: research agenda. J. Interact. Advert. 21 (3), 141–144. https://doi.org/10.1080/15252019.2021.2001273.

Kim, K., Yang, H., Lee, J., Lee, W.G., 2023. Metaverse wearables for immersive digital healthcare: a review. Adv. Sci. 10 (31), 2303234. https://doi.org/10.1002/advs.202303234.

Kock, N., 2015. Common method bias in PLS-SEM: a full collinearity assessment approach. Int. J. e-Collaboration 11 (4), 1–10. https://doi.org/10.4018/ijec.2015100101.

Koohang, A., Nord, J.H., Ooi, K.B., Tan, G.W.H., Al-Emran, M., Aw, E.C.X., et al., 2023. Shaping the metaverse into reality: a holistic multidisciplinary understanding of opportunities, challenges, and avenues for future investigation. J. Comput. Inf. Syst. 63 (3), 735–765. https://doi.org/10.1080/08874417.2023.2165197.

Kulikovskaja, V., Hubert, M., Grunert, K.G., Zhao, H., 2023. Driving marketing outcomes through social media-based customer engagement. J. Retailing Consum. Serv. 74, 103445. https://doi.org/10.1016/j.jretconser.2023.103445.

Kumar, A., Shankar, A., Agarwal, R., Agarwal, V., Alzeiby, E.A., 2024. With enterprise metaverse comes great possibilities! Understanding metaverse usage intention from an employee perspective. J. Retailing Consum. Serv. 78, 103767. https://doi.org/10.1016/j.jretconser.2024.103767.

Kumar, V., Rajan, B., Gupta, S., Pozza, I.D., 2019. Customer engagement in service. J. Acad. Market. Sci. 47, 138–160. https://doi.org/10.1007/s11747-017-0565-2.

Kye, B., Han, N., Kim, E., Park, Y., Jo, S., 2021. Educational applications of metaverse: possibilities and limitations. J. Educat. Evaluat. Health Professions 18, 32. https://doi.org/10.3352/jeehp.2021.18.32.

Lee, L.H., Braud, T., Zhou, P.Y., Wang, L., Xu, D., Lin, Z., et al., 2024. All one needs to know about metaverse: a complete survey on technological singularity, virtual ecosystem, and research agenda. Foundat. Trends® Human-Computer Interact. 18 (2–3), 100–337. http://arxiv.org/abs/2110.05352.

Lindner, J.R., Murphy, T.H., Briers, G.E., 2001. Handling nonresponse in social science research. J. Agric. Educ. 42 (4), 43–53. https://doi.org/10.5032/jae.2001.04043.

Lowe, B., 2010. Consumer perceptions of extra free product promotions and discounts: the moderating role of perceived performance risk. J. Prod. Brand Manag. 19 (7), 496–503. https://doi.org/10.1108/10610421011086919.

Malodia, S., Ferraris, A., Sakashita, M., Dhir, A., Gavurova, B., 2022. Can Alexa serve customers better? AI-driven voice assistant service interactions. J. Serv. Market. 37 (1), 25–39. https://doi.org/10.1108/jsm-12-2021-0488.

Man, S.S., Xiong, W., Chang, F., Chan, A.H., 2020. Critical factors influencing acceptance of automated vehicles by Hong Kong drivers. IEEE Access 8, 109845–109856. https://doi.org/10.1109/access.2020.3001929.

Marsh, H.W., Hocevar, D., 1985. Application of confirmatory factor analysis to the study of self-concept: first- and higher order factor models and their invariance across groups. Psychol. Bull. 97 (3), 562–582. https://doi.org/10.1037/0033-2909.97.3.562.

Mckinsey's, Hazan, Eric, Kelly, Greg, Khan, Hamza, Spillecke, Dennis, Yee, Lareina, 2022. Marketing in the metaverse: an opportunity for innovation and experimentation. Retrieved May 6, 2024, from. https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/marketing-in-the-metaverse-an-opportunity-for-innovation-and-experimentation.

McLean, G., Osei-Frimpong, K., 2019. Hey Alexa… examine the variables influencing the use of artificial intelligent in-home voice assistants. Comput. Hum. Behav. 99, 28–37. https://doi.org/10.1016/j.chb.2019.05.009.

Mehrotra, A., Agarwal, R., Khalil, A., Alzeiby, E.A., Agarwal, V., 2024. Nitty-gritties of customer experience in metaverse retailing. J. Retailing Consum. Serv. 79, 103876. https://doi.org/10.1016/j.jretconser.2024.103876.

Miles, M.B., 1994. Qualitative Data Analysis: an Expanded Sourcebook. Thousand Oaks.

Mitchell, V.W., 1999. Consumer perceived risk: conceptualisations and models. Eur. J. Market. 33 (1/2), 163–195. https://doi.org/10.1108/03090569910249229.
Mogaji, E., Balakrishnan, J., Nwoba, A.C., Nguyen, N.P., 2021. Emerging-market consumers' interactions with banking chatbots. Telematics Inf. 65, 101711. https://doi.org/10.1016/j.tele.2021.101711.
Mogaji, E., Wirtz, J., Belk, R.W., Dwivedi, Y.K., 2023. Immersive time (ImT): conceptualizing time spent in the metaverse. Int. J. Inf. Manag. 72, 102659. https://doi.org/10.1016/j.ijinfomgt.2023.102659.
Möhlmann, M., 2015. Collaborative consumption: determinants of satisfaction and the likelihood of using a sharing economy option again. J. Consum. Behav. 14 (3), 193–207. https://doi.org/10.1002/cb.1512.
Mollen, A., Wilson, H., 2010. Engagement, telepresence and interactivity in online consumer experience: reconciling scholastic and managerial perspectives. J. Bus. Res. 63 (9–10), 919–925. https://doi.org/10.1016/j.jbusres.2009.05.014.
Nadeem, W., Andreini, D., Salo, J., Laukkanen, T., 2015. Engaging consumers online through websites and social media: a gender study of Italian Generation Y clothing consumers. Int. J. Inf. Manag. 35 (4), 432–442. https://doi.org/10.1016/j.ijinfomgt.2015.04.008.
Natarajan, T., Pragha, P., Dhalmahapatra, K., 2024. Uses and gratifications of metaverse: understanding the user adoption factors through a mixed method approach. Asia Pac. J. Mark. Logist. https://doi.org/10.1108/APJML-03-2024-0395.
Nunkoo, R., Pillai, R., Sivathanu, B., Rana, N.P., 2024. Drivers and inhibitors of consumers' adoption of AI-driven drone food delivery services. Int. J. Hospit. Manag. 123, 103913. https://doi.org/10.1016/j.ijhm.2024.103913.
Ooi, K.B., Tan, G.W.H., Aw, E.C.X., Cham, T.H., Dwivedi, Y.K., Dwivedi, R., et al., 2023. Banking in the metaverse: a new frontier for financial institutions. Int. J. Bank Market. 41 (7), 1829–1846. https://doi.org/10.1108/IJBM-03-2023-0168.
Opazo-Basáez, M., Vendrell-Herrero, F., Bustinza, O.F., 2022. Digital service innovation: a paradigm shift in technological innovation. J. Serv. Manag. 33 (1), 97–120. https://doi.org/10.1108/JOSM-11-2020-0427.
Pantano, E., Rese, A., Baier, D., 2017. Enhancing the online decision-making process by using augmented reality: a two country comparison of youth markets. J. Retailing Consum. Serv. 38, 81–95. https://doi.org/10.1016/j.jretconser.2017.05.011.
Park, H., Lim, R.E., 2023. Fashion and the metaverse: clarifying the domain and establishing a research agenda. J. Retailing Consum. Serv. 74, 103413. https://doi.org/10.1016/j.jretconser.2023.103413.
Parasuraman, A., 2000. Technology readiness index (Tri). J. Serv. Res. 2 (4), 307–320. https://doi.org/10.1177/109467050024001.
Parasuraman, A., Colby, C.L., 2015. An updated and streamlined technology readiness index. J. Serv. Res. 18 (1), 59–74. https://doi.org/10.1177/1094670514539730.
Parasuraman, A., Grewal, D., 2000. The impact of technology on the quality-value-Loyalty chain: a research agenda. J. Acad. Market. Sci. 28 (1), 168–174. https://doi.org/10.1177/0092070300281015.
Parker, P.A., Janicki, R., Holan, S.H., 2019. Unit level modeling of survey data for small area estimation under informative sampling: a comprehensive overview with extensions. arXiv preprint arXiv:1908.10488. https://doi.org/10.48550/arXiv.1908.10488.
Payal, R., Sharma, N., Dwivedi, Y.K., 2024. Unlocking the impact of brand engagement in the metaverse on Real-World purchase intentions: analyzing Pre-Adoption behavior in a futuristic technology platform. Electron. Commer. Res. Appl. 65, 101381. https://www.sciencedirect.com/science/article/pii/S1567422324000267.
Pillai, R., Sivathanu, B., 2020. Adoption of artificial intelligence (AI) for talent acquisition in IT/ITeS organizations. Benchmark Int. J. 27 (9), 2599–2629. https://doi.org/10.1108/BIJ-04-2020-0186.
Pillai, R., Sivathanu, B., Rana, N.P., Struweg, I., 2024. Assessing the determinants of metaverse adoption for e-commerce retailing. J. Comput. Inf. Syst. 1–20. https://doi.org/10.1080/08874417.2024.2416452.
Podsakoff, P.M., MacKenzie, S.B., Lee, J.Y., Podsakoff, N.P., 2003. Common method biases in behavioral research: a critical review of the literature and recommended remedies. J. Appl. Psychol. 88 (5), 879–903. https://doi.org/10.1037/0021-9010.88.5.879.
Ramayah, T., Mohamad, O., Omar, A., Marimuthu, M., Leen, J.Y.A., 2013. Determinants of technology adoption among Malaysian SMEs: an IDT perspective. J. Informat. Commun. Technol. 12, 103–119.
Rasool, A., Shah, F.A., Islam, J.U., 2020. Customer engagement in the digital age: a review and research agenda. Curr. Opin. Psychol. 36, 96–100. https://doi.org/10.1016/j.copsyc.2020.05.003.
Rogers, E.M., 2003. Diffusion of Innovations, fifth ed. Free Press, New York.
Rodríguez-Ardura, I., Meseguer-Artola, A., Herzallah, D., Fu, Q., 2025. Pumping up customer value with convenience and personalisation strategies in e-retailing: an analysis of the engagement connection. J. Res. Indian Med. 19 (1), 35–58. https://doi.org/10.1108/JRIM-03-2023-0083.
Roy, S.K., Shekhar, V., Quazi, A., Quaddus, M., 2020. Consumer engagement behaviors: do service convenience and organizational characteristics matter? J. Serv. Theory Pract. 30 (2), 195–232. https://doi.org/10.1108/jstp-03-2018-0049.
Ryan, J., Casidy, R., 2018. The role of brand reputation in organic food consumption: a behavioral reasoning perspective. J. Retailing Consum. Serv. 41, 239–247. https://doi.org/10.1016/j.jretconser.2018.01.002.
Sagnier, C., Loup-Escande, E., Lourdeaux, D., Thouvenin, I., Valléry, G., 2020. User acceptance of virtual reality: an extended technology acceptance model. Int. J. Hum. Comput. Interact. 36 (11), 993–1007. https://doi.org/10.1080/10447318.2019.1708612.
Sahu, A.K., Padhy, R., Dhir, A., 2020. Envisioning the future of behavioral decision-making: a systematic literature review of behavioral reasoning theory. Australas. Market J. 28 (4), 145–159. https://doi.org/10.1016/j.ausmj.2020.05.001.
Shahzad, K., Ashfaq, M., Zafar, A.U., Basahel, S., 2024. Is the future of the metaverse bleak or bright? Role of realism, facilitators, and inhibitors in metaverse adoption. Technol. Forecast. Soc. Change 209, 123768. https://doi.org/10.1016/j.techfore.2024.123768.
Shen, V., Rae-Grant, T., Mullenbach, J., Harrison, C., Shultz, C., 2023. Fluid reality: high-resolution, untethered haptic gloves using electroosmotic pump arrays. In: Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology, pp. 1–20. https://doi.org/10.1145/3586183.3606771.
Shukla, A., Mishra, A., Rana, N.P., Banerjee, S., 2024. The future of metaverse adoption: a behavioral reasoning perspective with a text-mining approach. J. Consum. Behav. 23 (5), 2217–2233. https://doi.org/10.1002/cb.2336.
Silva, S.C., Rocha, T.V., De Cicco, R., Galhanone, R.F., Mattos, L.T.M.F., 2021. Need for touch and haptic imagery: an investigation in online fashion shopping. J. Retailing Consum. Serv. 59, 102378. https://doi.org/10.1016/j.jretconser.2020.102378.
Singla, B., Bhattacharya, S., Naik, N., 2023. Introduction to metaverse and consumer Behaviour change: adoption of metaverse among consumers. In: Handbook of Research on Consumer Behavioral Analytics in Metaverse and the Adoption of a Virtual World. IGI Global, pp. 113–129. https://doi.org/10.4018/978-1-6684-7029-9.ch006.
Smith, S., Paladino, A., 2010. Eating clean and green? Investigating consumer motivations towards the purchase of organic food. Australas. Mark. J. 18, 93–104. https://doi.org/10.1016/j.ausmj.2010.1001.1001.
Stone, R.N., Grønhaug, K., 1993. Perceived risk: further considerations for the marketing discipline. Eur. J. Market. 27 (3), 39–50. https://doi.org/10.1108/03090569310026637.
Sun, Q., Spiegelman, D., van Dam, R.M., Holmes, M.D., Malik, V.S., Willett, W.C., Hu, F. B., 2010. White rice, brown rice, and risk of type 2 diabetes in US men and women. Arch. Intern. Med. 170 (11), 961–969. https://doi.org/10.1001/archinternmed.2010.109.
Sweeney, J.C., Soutar, G.N., Johnson, L.W., 1999. The role of perceived risk in the quality-value relationship: a study in a retail environment. J. Retailing 75 (1), 77–105. https://doi.org/10.1016/S0022-4359(99)80005-0.
Tandon, A., Dhir, A., Kaur, P., Kushwah, S., Salo, J., 2020. Behavioral reasoning perspectives on organic food purchase. Appetite 154, 104786. https://doi.org/10.1016/j.appet.2020.104786.
Taylor, A., Hook, M., Carlyle, T., Carlson, J., 2024. Creating a metaverse-me: exploring the consumer avatar creation process. J. Consum. Behav. 23 (6), 2846–2861.
Tlili, A., Huang, R., Shehata, B., Liu, D., Zhao, J., Metwally, A.H.S., et al., 2022. Is Metaverse in education a blessing or a curse: a combined content and bibliometric analysis. Smart Learn. Environ. 9 (1), 1–31. https://doi.org/10.1186/s40561-022-00205-x.
Trivedi, J., Kasilingam, D., Arora, P., Soni, S., 2022. The effect of augmented reality in mobile applications on consumers' online impulse purchase intention: the mediating role of perceived value. J. Consum. Behav. 21 (4), 896–908. https://doi.org/10.1002/cb.2047.
Van Doorn, J., Lemon, K.N., Mittal, V., Nass, S., Pick, D., Pirner, P., Verhoef, P.C., 2010. Customer engagement behavior: theoretical foundations and research directions. J. Serv. Res. 13 (3), 253–266. https://doi.org/10.1177/1094670510375599.
Van Esch, P., Black, J.S., Ferolie, J., 2019. Marketing AI recruitment: the next phase in job application and selection. Comput. Hum. Behav. 90, 215–222. https://doi.org/10.1016/j.chb.2018.09.009.
Van, H.N., Pham, L., Williamson, S., Huong, V.T., Hoa, P.X., Trang, P.L.H., 2020. Impact of perceived risk on mobile banking usage intentions: trust as a mediator and a moderator. Int. J. Bus. Emerg. Mark. 12 (1), 94–118. https://doi.org/10.1504/IJBEM.2020.106202.
Vimalkumar, M., Sharma, S.K., Singh, J.B., Dwivedi, Y.K., 2021. 'Okay google, what about my privacy?': user's privacy perceptions and acceptance of voice based digital assistants. Comput. Hum. Behav. 120, 106763. https://doi.org/10.1016/j.chb.2021.106763.
Venkatesh, V., Morris, M.G., Davis, G.B., Davis, F.D., 2003. User acceptance of information technology: toward a unified view. MIS Q. 425–478. https://doi.org/10.2307/30036540.
Venkatesh, V., Thong, J.Y., Xu, X., 2016. Unified theory of acceptance and use of technology: a synthesis and the road ahead. J. Assoc. Inf. Syst. Online 17 (5), 328–376. https://ssrn.com/abstract=2800121.
Vivek, S.D., Beatty, S.E., Morgan, R.M., 2012. Customer engagement: exploring customer relationships beyond purchase. J. Market. Theor. Pract. 20 (2), 122–146. https://doi.org/10.2753/MTP1069-6679200201.
Wasiq, M., Bashar, A., Khan, I., Nyagadza, B., 2024. Unveiling customer engagement dynamics in the metaverse: a retrospective bibliometric and topic modelling investigation. Comput. Human Behav. Rep. 16, 100483. https://doi.org/10.1016/j.chbr.2024.100483.
Waung, M., McAuslan, P., Lakshmanan, S., 2021. Trust and intention to use autonomous vehicles: manufacturer focus and passenger control. Transport. Res. F Traffic Psychol. Behav. 80, 328–340. https://doi.org/10.1016/j.trf.2021.05.004.
Westaby, J.D., 2005. Behavioral reasoning theory: identifying new linkages underlying intentions and behavior. Organ. Behav. Hum. Decis. Process. 98 (2), 97–120. https://doi.org/10.1016/j.obhdp.2005.07.003.
Williams, M., Moser, T., 2019. The art of coding and thematic exploration in qualitative research. Int. Manage. Rev. 15 (1), 45–55.
Xi, N., Chen, J., Gama, F., Korkeila, H., Hamari, J., 2024. Acceptance of the metaverse: a laboratory experiment on augmented and virtual reality shopping. Internet Res. 34 (7), 82–117. https://doi.org/10.1108/INTR-05-2022-0334.
Xu, F., Michael, K., Chen, X., 2013. Factors affecting privacy disclosure on social network sites: an integrated model. Electron. Commer. Res. 13, 151–168. https://doi.org/10.1007/s10660-013-9111-6.

Yale, L.J., Venkatesh, A., 1986. Toward the Construct of Convenience in Consumer Research. ACR North American Advances.
Yaqoob, I., Salah, K., Jayaraman, R., Omar, M., 2023. Metaverse applications in smart cities: enabling technologies, opportunities, challenges, and future directions. Internet Things, 100884. https://doi.org/10.1016/j.iot.2023.100884.
Yu, H., 2024. Why do people use Metaverse? A uses and gratification theory perspective. Telematics Inf. 89, 102110. https://doi.org/10.1016/j.tele.2024.102110.
Zallio, M., Clarkson, P.J., 2022. Designing the metaverse: a study on inclusion, diversity, equity, accessibility and safety for digital immersive environments. Telematics Inf. 75, 101909. https://doi.org/10.1016/j.tele.2022.101909.
Zhang, H., Lv, Y., Zhang, J.Z., Hollebeek, L.D., Behl, A., Urbonavicius, S., 2025. Exploring purchase intention in metaverse retailing: insights from an automotive platform. J. Retailing Consum. Serv. 82, 104144. https://doi.org/10.1016/j.jretconser.2024.104144.
Zhang, R.W., Liang, X., Wu, S.H., 2024. When chatbots fail: exploring user coping following a chatbots-induced service failure. Inf. Technol. People 37 (8), 175–195. https://www.emerald.com/insight/content/doi/10.1108/ITP-08-2023-0745/full/html.
Zhao, R., Zhang, Y., Wang, T., Wen, W., Xiang, Y., Cao, X., 2023a. Visual content privacy protection: a survey. arXiv preprint arXiv:2303.16552. https://arxiv.org/abs/2303.16552.
Zhao, R., Zhang, Y., Zhu, Y., Lan, R., Hua, Z., 2023b. Metaverse: security and privacy concerns. J. Metaverse 3 (2), 93–99. https://doi.org/10.57019/jmv.1286526.
Zhou, X., Yan, X., Jiang, Y., 2024. Making sense? The sensory-specific nature of virtual influencer effectiveness. J. Market. 88 (4), 84–106. https://doi.org/10.1177/00222429231203699.